Report R32 — Floquet spectra and thermalisation of the sixteen unitary quantum cellular automata

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

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Abstract

We study whether the dynamics inside the Krylov sectors of the sixteen unitary controlled-Hadamard brickwork circuits (the quantum elementary cellular automata of Hillberry *et al.*) thermalise. The tools are symmetry-resolved quasienergy statistics of the one-period operator U, eigenstate entanglement, the spectral form factor, ETH for a local observable and quench dynamics. Open chains reach $N = 14$ to 20, depending on the rule; rings reach $N = 16$.

- *One number organises the rules:* the count k of firing neighbour patterns.
- $k = 3$ (54, 57, 99, 147) is chaotic on both boundaries, with COE or CUE as the symmetry dictates.
- *The one-sided $k = 2$ rules* are chaotic on rings, protected by a newly identified dressed-reflection time reversal, but form a non-ergodic tower of conserved boundary projectors on open chains.
- *XOR* (free fermions) and $k = 1$ (exponential fragmentation) do not thermalise.
- *All exact scars* of the chaotic rules descend from one universal eigenstate, $|h_+\rangle^{\otimes N}$.
- *With the X gate* the chaotic rule 54 is the integrable RCA54: quantum chaos follows the firing count, not the classical dynamics.

Sec. 1 summarises all results, together with what was known before and what is new.

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1 Summary

1.1 Question and model

The Krylov analysis of the earlier reports describes *how* the Hilbert space of the sixteen unitary rules falls apart. This report asks whether the dynamics *inside* the pieces thermalise.

The circuits are the quantum elementary cellular automata of Hillberry and coworkers [37, 38] (Sec. 3.5):

- On a chain of N qubits, the Hadamard H acts on site j iff the neighbour pattern (x_{j-1}, x_{j+1}) is one of the rule's *firing patterns*.
- Even sites are updated first, then odd sites (brickwork), and one period is $U = L_B L_A$.
- Two boundaries are studied: open with ghost neighbours fixed to 0 (obc0), and rings of even length.
- The X gate turns every rule into a classical reversible automaton and serves as the comparison.

The tools are applied to U itself, not to a Hamiltonian:

- the quasienergy gap ratio $\langle r \rangle$ after *full* symmetry resolution (Krylov sectors, gate-algebra blocks, reflection or the half step Ω on the chain, and momentum with the half step Ω_T on the ring);
- the eigenstate entanglement against the Page value;
- the spectral form factor;
- ETH for a local observable;
- quench dynamics.

1.2 The classification

One combinatorial number organises the sixteen rules: the count k of the four neighbour patterns that fire the gate, and, for $k = 2$, whether the firing set depends on one neighbour or on both (Table 1).

class	rules	open chain	ring
$k = 3$	54, 57, 99, 147	**chaotic**: $\langle r \rangle = 0.523$–$0.531$ at $N = 14$ (GOE 0.531)	**chaotic**: 57/99 CUE (0.5985 at $N = 16$), 147 COE (0.528), 54 COE but slow (0.510, rising)
$k = 2$ one-sided	60, 102, 153, 195	tower of conserved boundary projectors; $\langle r \rangle \approx 0.424$, flat for $N = 13 \ldots 16$	**chaotic**, COE (0.523–0.527 at $N = 16$), protected by a hidden time reversal
$k = 2$ XOR	105, 150	arithmetic: exactly 3^m distinct levels for 150 ($N = 6 \ldots 15$)	arithmetic (150 is free fermions)
$k = 1$	108, 156, 198, 201	exponential fragmentation; 108/201 Poisson, 156/198 $\langle r \rangle \approx 0.42$ (common $N = 19$)	unchanged
$k = 0$, 4	204, 51	trivial ($U = 1$, $U = H^{\otimes N}$)	trivial

Table 1: The sixteen rules with $V = H$. Each class is compared at one common N (Sec. 6); ring values are for the complex momentum sectors (Sec. 7).

Only the $k = 3$ class is chaotic on both boundaries. Its ensemble follows the symmetry. With open boundaries every block carries the time reversal KL_A (COE). On the ring, the mirror rules 54 and 147 keep a time reversal KL_AR (COE). 57 and 99 have none inside a momentum sector, so they are CUE, with a GUE-shaped $P(r)$ (Sec. 7). Without symmetry resolution, 147 looks integrable (0.42 against 0.53).

The one-sided class is a boundary phenomenon.

- *Open chain:* the free end carries a chain of conserved projectors onto Hadamard eigenstates, which builds a tower of blocks of size $2^{N-1}, 2^{N-2}, \ldots$. Inside the largest block the statistics plateau at $\langle r \rangle \approx 0.424$ for $N = 13 \ldots 16$: neither Poisson nor GOE, and not drifting.
- *Ring:* there is no tower and the rules are COE-chaotic, although they have no mirror symmetry. The time reversal is a newly identified *dressed bond reflection*: the reflection is combined with a rotation of one sublattice into the Hadamard eigenbasis (Sec. 10).
- *Dynamics:* random product states reach only 0.72 (60) and 0.91 (153) of the Page entanglement on the open chain, but 0.99 to 1.00 on the ring.

XOR and $k = 1$ do not thermalise. 150 with H is a free-fermion circuit [40]; its arithmetic spectrum reproduces R8's level count exactly. The $k = 1$ rules split into exponentially many small blocks.

- In 156/198 a block is a product of segments, each holding one mobile kink with the exact dispersion $\cos\theta = -\sin^2(\pi m/(L+1))$. The sums of these few levels give $\langle r \rangle \approx 0.42$.
- 108/201 have larger, many-body rooms and Poisson statistics.

1.3 Thermalisation

All diagnostics give the same ordering (Sec. 9, Table 6):

- *57/99 thermalise fastest and most cleanly:* ETH fluctuations decay as $d^{-0.40}$, close to $d^{-1/2}$; the SFF approaches the ramp after $t_2 = 78$ periods at $N = 16$; random states reach Page.
- *153, 147, 60 and 54* are chaotic on the ring but converge progressively more slowly. Their ETH exponents are -0.24 to -0.13, t_2 is 5 to 13 times longer, and the SFF has a large short-time excess.
- *XOR and $k = 1$* stay far from thermal in every diagnostic.

The gap ratio is the most robust single diagnostic, but not a sufficient one: the chaotic rules contain non-thermal eigenstates and slow states (next subsection).

1.4 Exact structures behind the classes

- **One universal scar.** $|h_+\rangle^{\otimes N}$, the product of $+1$ Hadamard eigenstates, is an eigenstate of *every* controlled-Hadamard gate, for every rule and boundary. It explains all exact scars of the chaotic rules:
  - the state itself ($\theta = 0$);
  - its projection off a frozen vacuum (the entangled singlet of 54, 60 and 102, with entanglement in closed form, decreasing with N);

– $|h_-\rangle$ on boundary sites whose gate is control-independent ($\theta = \pi$; the counts 2/2/4/1 follow exactly).

Close to this vacuum live the quasi-bound edge mode of 57 (quasienergy $\pm 0.312\pi$) and the slowly relaxing defect states of 54 and 147.

- **Flip partners.** $X^{\otimes N} U_W(H) X^{\otimes N} = U_{W'}(XHX) \cong U_{W'}(-H)$: the partner rule with a -1 on every firing. This explains why flip partners share their block structure but not their spectra, and the obc0 boundary breaks the flip even for X.

- **The X gate.** Classically, 54/147 is the integrable RCA54, 201/108 is the Floquet-PXP automaton, and XOR and the one-sided rules are linear. X-102 is exactly the solvable deterministic Floquet quantum East model [44]. Only 57/99 have orbits that fill a finite fraction of the space, and only on the open chain; on the ring a conserved domain-wall number confines them. *Quantum chaos with H therefore follows the firing count, not the classical dynamics*: the integrable RCA54 becomes a chaotic circuit with H.

1.5 What is known and what is new

Known:

- the model and its numbering [37, 38];
- entanglement-spectrum (Brody) indications that T_{13} and T_{14} (our 57, 54) are chaotic and T_1 (201) is not [38];
- the free-fermion nature and the domain-wall charge of the Hadamard Goldilocks rule 150 [40];
- the classical integrable parents, and the solvable DFQEM (= X-102) [33, 34, 44].

New, as far as we found (Sec. 3.5):

- the symmetry-resolved quasienergy classification of all sixteen rules on both boundaries, organised by k;
- the boundary origin of the one-sided class and its hidden time reversal;
- the universal scar and its consequences;
- the exact kink dispersion of 156/198;
- the flip map;
- the domain-wall charge of X-57;
- the SFF, ETH and quench characterisation of all classes.

1.6 Open questions

1. Why the one-sided rules plateau at $\langle r \rangle \approx 0.424$ on the open chain. The tower block is the same chain with one boundary site fixed in $|h_-\rangle$; some further structure seems to be at work.
2. The origin of the large short-time SFF excess and the slow ETH convergence of 54, 60 and 147, presumably tied to the $|h_+\rangle$ vacuum and its slow defects.
3. An analytic form of the 57 edge mode.
4. Whether 54 reaches COE on the ring (0.510 at $N = 16$, still rising).

Reading guide.

- Sec. 3: the tools for Floquet operators and the literature on these circuits.
- Secs. 4–5: methods and the optimised pipeline.
- Sec. 6: open-chain results.
- Sec. 7: rings.
- Sec. 8: the X gate.
- Sec. 9: thermalisation.
- Sec. 10: the exact structures and the runs beyond the dense limit.
- Sec. 11: compute.

2 Scope and history

Task (2026-10-14):

1. review the literature on eigenstate entanglement, level statistics and scars for Floquet operators;
2. complete the spectral dataset for the 16 unitary rules;
3. per-rule figures;
4. interpretation;
5. a storyline draft and candidate thesis figures, to be shown before any hand-off;
6. the same for the X gate (optional).

Requests added during the work:

- resolve all symmetries;
- check the classes on rings;
- compare each class at one common N;
- resolve the open physics points and push past the dense-sector limit for the relevant cases, verifying every assumption;
- add the spectral form factor, ETH and dynamics;
- a second literature pass (Hillberry; the Floquet quantum East circuits).

Corrections made along the way are recorded where they occur (Secs. 7, 10) and in `NEXT_STEPS.md`.

3 The tools, and what they mean for a Floquet operator

Everything in this report is a property of one operator, the one-period evolution $U = L_B L_A$ of the brickwork circuit, and not of a Hamiltonian. This section collects what the literature says the three diagnostics mean in that setting, and where the Hamiltonian intuition has to be dropped.

3.1 Quasienergies, and why there is no "middle of the spectrum"

For a time-periodic Hamiltonian $H(t+T) = H(t)$ or a periodically applied circuit, the stroboscopic dynamics is generated by the Floquet operator $U = \mathcal{T}\exp[-i\int_0^T H(t)\,dt]$, here simply the product of the two gate layers. Its eigenvalues lie on the unit circle, $U\,|\phi_n\rangle = e^{-i\theta_n}\,|\phi_n\rangle$, and the quasienergies θ_n are defined only modulo 2π. There is no ordering, no ground state and no temperature axis.

For a generic interacting driven system the long-time state is the *infinite-temperature* state. D'Alessio and Rigol showed by exact diagonalisation and finite-size scaling that for any non-zero driving period the Floquet operator of a generic lattice system acquires circular-ensemble properties as the system grows. The Floquet Hamiltonian is then non-local and the system heats to infinite temperature [8]. Lazarides, Das and Moessner reached the same conclusion from the structure of Floquet eigenstates, which mix many eigenstates of the undriven Hamiltonian. They also showed that integrable driven systems are instead described by a periodic generalised Gibbs ensemble [9]. Ponte, Chandran, Papić and Abanin found that ergodic systems delocalise in energy space and heat up under both local and global driving [10]. Disorder can prevent this: the Floquet eigenstates of a many-body localised system obey an area law, and a delocalised phase is separated from the MBL phase by a transition seen in the eigenstate entanglement [11].

Consequence for the figures. The Floquet version of the eigenstate thermalisation hypothesis (ETH) [12, 13, 14, 15] says that every eigenstate of a chaotic U looks locally like the infinite-temperature state. Plotted against θ, the eigenstate entanglement of a thermalising circuit is therefore *flat*, a band at the random-state value at every quasienergy. There is no Hamiltonian "dome" that peaks in the middle of the spectrum. A non-thermal (scar-like) eigenstate shows up as an outlier *below* the band, at whatever quasienergy it happens to sit.

3.2 Spectral statistics: circular ensembles and the gap ratio

Dyson's circular ensembles are the random-matrix models for unitary operators [3]: orthogonal (COE, $\beta = 1$), unitary (CUE, $\beta = 2$) and symplectic (CSE, $\beta = 4$). The Bohigas–Giannoni–Schmit conjecture [4] assigns random-matrix level statistics to quantum chaotic systems. The Berry–Tabor conjecture assigns Poisson statistics to integrable ones, with uncorrelated, clustering levels [5]. Which circular ensemble applies is fixed by the antiunitary symmetries *of U itself*. This need not match the class of the instantaneous Hamiltonian: Regnault and Nandkishore showed that a Floquet system can be in the unitary class while its instantaneous Hamiltonian is always orthogonal, and vice versa. They also showed that a drive can thermalise even when the instantaneous Hamiltonian is integrable [6].

The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ of consecutive spacings, introduced by Oganesyan and Huse [2], is independent of the local density of states and so needs no unfolding. Atas *et al.* derived Wigner-like surmises for its distribution

$$P_\beta(r) = \frac{2}{Z_\beta}\,\frac{(r+r^2)^\beta}{(1+r+r^2)^{1+3\beta/2}}, \qquad r \in [0,1], \tag{1}$$

with $Z_1 = 8/27$ and $Z_2 = 4\pi/(81\sqrt{3})$. Their large-matrix values, $\langle r\rangle = 0.5307$ (GOE) and 0.5996 (GUE), are the reference values used here. Poisson statistics give $P_0(r) = 2/(1+r)^2$ and $\langle r\rangle = 2\ln 2 - 1 = 0.3863$ [1]. For a chaotic Floquet operator the quasienergy density is uniform, so the ratio is doubly convenient.

Two caveats from the literature are central to this report:

- **Symmetry sectors must be resolved.** Superposing independent spectra from different symmetry sectors removes level repulsion between them and pulls $\langle r\rangle$ towards the Poisson value, even if every sector is chaotic. Giraud, Macé, Vernier and Alet derived the gap-ratio surmises for such superpositions and used them to detect hidden symmetries,

including in periodically driven spin chains [7]. ETH, too, is a statement per symmetry sector. Mondaini *et al.* made exactly this point in their comment [21] on a claimed ETH counterexample [20].

- **In fragmented systems the Krylov sectors play the role of symmetry sectors.** Hilbert-space fragmentation (HSF) splits the space into exponentially many invariant subspaces that are not labelled by conventional symmetries [26, 27]. In the commutant language of Moudgalya and Motrunich [28], the invariant subspaces are the blocks of the algebra generated by the local terms (or gates). Its commutant has exponential dimension, and the product-basis Krylov sectors ("classical" fragmentation) can be refined further by quantum fragmentation [28]. Level statistics are meaningful only inside one such block. Pooling blocks measures their number, not their chaoticity.

3.3 Eigenstate entanglement

For a random pure state on $\mathbb{C}^m \otimes \mathbb{C}^n$ with $m \leq n$, Page's mean entanglement entropy is $S_{\text{Page}} \approx \ln m - m/(2n)$, short of the maximum by about half a unit of information [16]. Eigenstates of chaotic Hamiltonians approach this value in the middle of the spectrum. Conservation laws lower it: Vidmar and Rigol showed that for random states at fixed particle number away from half filling, the average entropy falls short of the maximum by an amount growing as the square root of the volume [17]. For a fragmented system the right benchmark is therefore a random state *inside the relevant block*, not S_{Page} of the full space.

Volume-law eigenstates signal thermalisation. Area-law or otherwise anomalously weakly entangled eigenstates embedded in a thermal spectrum are the defining signature of quantum many-body scars (QMBS). QMBS were identified as the origin of the persistent revivals in the Rydberg-atom experiment [18], via scarred eigenstates of the PXP model [19]. Embedding constructions produce them systematically [20]. Reviews cover the semiclassical picture [22], exact results [23], and the quasiparticle perspective [24]. Scars also exist intrinsically in driven systems: Sugiura, Kuwahara and Saito proved that a PXP-type Floquet model hosts exact scar eigenstates of U at specific driving periods, a rigorous breakdown of Floquet ETH [25].

Degeneracies make single-eigenvector entanglement ill-defined. If an eigenvalue of U is g-fold degenerate, any orthonormal basis of the eigenspace is equally valid, and the entanglement of "the" eigenvectors depends on that arbitrary choice. Earlier R8 found that 43–74 % of the stored rule-150 eigenstates lie in such eigenspaces, with multiplicity up to 35. This report therefore (i) resolves all symmetries first, so the remaining basis freedom is as small as possible, (ii) records g for every state, and (iii) adds a basis-independent quantity for every eigenspace: the Haar-averaged Rényi-2 entropy (Sec. 4).

3.4 Quantum circuits and automata in particular

Brickwork Floquet circuits are the setting in which many-body quantum chaos has been connected analytically to random-matrix theory. Kos, Ljubotina and Prosen derived RMT spectral correlations for periodically kicked many-body systems [29]. Chan, De Luca and Chalker solved a Floquet circuit of random two-site gates (depth two, like ours) in the large-q limit, including its spectral form factor and entanglement growth [30]. The circuits studied here are the opposite extreme: *one fixed* controlled gate, repeated.

Automaton-based circuits are a known source of non-ergodic eigenstates. Gopalakrishnan and Zakirov constructed facilitated quantum cellular automata, mostly from Clifford gates. Their eigenstate entanglement differs from generic chaotic systems in all five models, and some models have exponentially many eigenstates with disentangled sites, tied to reflection symmetry [31]. Iadecola and Vijay showed that deforming a classical reversible cellular automaton gives generically non-integrable Floquet systems that inherit exact non-thermal eigenstates from the

automaton's periodic orbits [32]. The automata themselves (our $V = X$ circuits) include the interacting integrable Rule 54 [33] and the Floquet-PXP Rule 201, whose solitons are domain walls between three vacua [34].

3.5 What is known about these sixteen circuits

The model is Hillberry's quantum elementary cellular automaton. Our circuits are exactly the "quantum elementary cellular automata" (QECA) introduced by Hillberry in his 2016 Master's thesis [37]. They are the "T-type" rules of Hillberry *et al.* [38]:

- a single-qubit unitary V acts on site j iff the neighbour pattern (m, n) is selected by the rule number $S = 8s_{11} + 4s_{10} + 2s_{01} + s_{00}$, which is our HSF numbering, so T_S is our HSF rule S;
- even sites are updated first, then odd sites (their "alt" mode, our brickwork);
- the boundary neighbours are fixed to $|0\rangle$ (our obc0);
- the main choice of gate is $V = HP(\theta)$ with phase angle $\theta = 0$, i.e. $V = H$.

The dictionary is $T_1 = 201$, $T_6 = 150$, $T_{13} = 57$ and $T_{14} = 54$.

What those works established.

- *Hillberry 2016 [37]:* exact dynamics from single-excitation initial states at $L \leq 27$: bond entropy, mutual-information networks, Fourier spectra of $\langle \sigma_j^z \rangle$, and transport. Rule 6 (our 150) is singled out as uniquely "complex", with persistent, slowly equilibrating entanglement. Rules 5, 7 and 11 (our 153, 147, 99) grow entanglement fast. There is no spectral or symmetry analysis.
- *Hillberry et al. 2021 [38]:* the "Goldilocks" rule T_6 (our 150) generates complexity (mutual-information networks, persistent entropy fluctuations, and Page curves closer to an area law). A Brody parameter fitted to *entanglement spectra* of states evolved from one $|1\rangle$ ($L = 19$) reaches chaotic values for T_{13} and T_{14} (our 57 and 54) and stays Poissonian for T_1 (our 201). For T_6 it "defies the paradigm" (many eigenvalues ≈ 0). The authors state that QECA "defy the chaos/integrability classification". T_6 was later run on up to 23 superconducting qubits [39].
- *Hillberry, Piroli, Vernier, Yunger Halpern, Prosen and Carr [40]:* the Goldilocks (XOR-controlled) circuits conserve the number of domain walls $Q_1 = \sum_j \sigma_j^z \sigma_{j+1}^z$ for any V. They are *free fermions*, proved both by Jordan–Wigner and by a six-vertex mapping, for $V_{\text{free}}(\alpha, \beta, \pm)$. This family contains the Hadamard ($\alpha = \pi/4$, $\beta = 0$, $-$ sign), i.e. the experimentally implemented T_6 and our 150. For generic V, symmetry-resolved quasienergy statistics on a ring ($L = 16$, sectors of momentum and Q_1) approach Wigner–Dyson.
- *Classical parents ($V = X$):* RCA54 is the interacting integrable Rule 54 [33], and 201 is the Floquet-PXP automaton [34]. Deforming a classical automaton by non-commuting gates gives chaotic Floquet circuits with exact non-thermal eigenstates on classical periodic orbits [32]. The PXP-type quantum automaton (the constraint of our 201) shows ergodicity breaking carried by chiral quasiparticles [42]. A different quantum deformation of RCA54 is integrable [41]: X is kept on the active patterns and only the 00 pattern carries a general $U(2)$ gate. It does not contain our 54, which acts with H on the active patterns and with the identity on 00.

- *One-sided constraint: the Floquet quantum East family.* In the deterministic Floquet quantum East model (DFQEM) a qubit flips iff its *right* neighbour is 1. The CNOT gates are laid out as a brickwork on a ring, so the DFQEM is exactly our *X-gate rule 102*, and 60 is its left-facilitated mirror image. Results:

  – Its quench dynamics are solved exactly [44]. Entanglement grows at half the speed allowed by locality, and blocks relax to infinite temperature. The solvability is tied to a hierarchical generalisation of dual unitarity [50].

  – The latest work builds on that solution. De Fazio, Lesanovsky and Perfetto compute the exact first-detection probability of a locally monitored region [49]. The monitored, stochastic and hardware versions show dynamical phase coexistence [45, 46, 48, 47].

  – With periodic boundaries the fully de-excited (vacuum) state decouples as an exceptional stationary state [48].

  – For the non-deterministic Floquet quantum East family $e^{-i\omega n\,\sigma^x}$, a disorder-free localisation transition exists [43].

  Our 102 and 60 with $V = H$ are a different deformation of the DFQEM, with the same constraint but a Hadamard instead of X, outside the $e^{-i\omega\sigma^x}$ family. None of these results applies to them directly. Two points of contact are new: the vacuum that decouples there is the frozen Krylov sector of our 60/102 on the ring, and our entangled scar of 60/102 is $|h_+\rangle^{\otimes N}$ projected off exactly that state.

What R32 adds. Measured against these works, the following results appear to be new; we found none of them in the literature above or in the works cited there.

1. A *symmetry-resolved quasienergy classification of all sixteen rules* with $V = H$, on both boundaries. It resolves Krylov sectors, gate-algebra blocks, reflection or the half step Ω (Ω_T on the ring), and momentum. The organising number is the firing count k. The earlier chaos/integrability statements for four rules rested on Brody fits to entanglement spectra of single trajectories; the quasienergy statistics confirm them where they overlap (T_{13}, T_{14} chaotic; T_1 not) and complete them.

2. The *boundary dependence* of the one-sided class: a tower of conserved Hadamard-basis projectors at the free end on the open chain, and COE chaos on the ring. The latter is protected by a newly identified time reversal, a dressed bond reflection.

3. The *exact scars* of the chaotic rules as one universal state, $|h_+\rangle^{\otimes N}$: its projection off a frozen vacuum (the entangled singlet of 54, 60 and 102, in closed form), and boundary $|h_-\rangle$ states, whose counts are predicted. Also the edge mode of 57 on that vacuum.

4. The *exact single-kink dispersion* of 156/198, and the explanation of their $\langle r\rangle \approx 0.42$.

5. The *flip map* $X^{\otimes N}U_W(H)X^{\otimes N} = U_{W'}(XHX) \cong U_{W'}(-H)$, which explains why flip partners share block structures but not spectra.

6. For the *X gate*: the domain-wall charge of X-57 on the ring, and the resulting boundary dependence of its orbit structure.

Known and only confirmed here are: the exact solvability of X-gate rule 102 (= the DFQEM [44]) and of its mirror 60, whose short, N-independent obc0 cycle lengths (16 or 32) are consistent with their linearity; the free-fermion (hence arithmetic) nature of the Hadamard Goldilocks rule 150, with our exact level counting 3^m consistent with it; its domain-wall charge; and the classical integrability of RCA54 and of Floquet-PXP.

3.6 Two structural facts used throughout

U is real orthogonal and a product of two reflections. Both gates, H and X, are real symmetric involutions, so every controlled gate is one and each layer $L_{A,B}$ is a real symmetric involution (a reflection). Hence $U = L_B L_A$ is real orthogonal. Two consequences:

1. Complex conjugation K commutes with U, so within every symmetry-resolved block the spectrum is symmetric under $\theta \to -\theta$. The independent spectral data are the $|\theta| \in [0, \pi]$. Ratios computed over the full circle would count each spacing twice and add spurious small gaps where mirror pairs approach 0 and π.

2. By the two-subspace theorem [35, 36], the space decomposes into the intersections of the ± 1 eigenspaces of L_A and L_B (eigenvalues ± 1 of U) and two-dimensional subspaces on which U rotates by $\pm 2\varphi_j$, where the φ_j are the principal angles between the $+1$ eigenspaces of the two layers. For generic reflections these angles follow a real ($\beta = 1$) Jacobi ensemble, whose bulk statistics are those of the GOE. We verified the reference value directly with the same routine used for the data: products of two Haar-random reflections of size 300–800 give $\langle r \rangle = 0.533 \pm 0.004$ on the half spectrum, compared with 0.535 ± 0.005 for a GOE control and 0.604 ± 0.005 for a GUE control.

In addition $T = L_A K$ is an antiunitary with $TUT^{-1} = U^{-1}$ and $T^2 = 1$, a time-reversal symmetry, and also points to the orthogonal class. The chaotic reference is therefore $\langle r \rangle_{\mathrm{GOE}} = 0.5307$ for every block that T maps to itself: all blocks with open boundaries, and the self-conjugate momentum sectors on a ring. In a ring-momentum sector $j \neq -j$, T maps the sector to its partner $-j$, no antiunitary symmetry remains inside it, and the chaotic reference is the CUE value 0.5996.

The X gate makes U a permutation. With $V = X$ every basis state is mapped to a single basis state. On a cycle of length ℓ the spectrum is the ℓ-th roots of unity, a picket fence with $r \equiv 1$, and the eigenvectors are Fourier modes along the cycle. Each Krylov sector of a permutation is a single cycle, so this holds sector by sector. The commutant study R-T26 found exactly this, $P(r) = \delta(r - 1)$, for rule 156. Neither Poisson nor RMT applies, and an X-gate $\langle r \rangle$ measures orbit arithmetic, not chaos.

4 Methods

4.1 Model and conventions

The sixteen unitary rules are the tuples $(r_{00}, r_{01}, r_{10}, r_{11}) \in \{I, V\}^4$, Wolfram numbers 51, 54, 57, 60, 99, 102, 105, 108, 147, 150, 153, 156, 195, 198, 201 and 204, with $V = H$ throughout this report (Sec. 8 for $V = X$). Boundary conditions are open with fixed vacuum neighbours (obc0). One period is $U = L_B L_A$, with L_A the gates on the odd (1-based) sites, applied first, and L_B those on the even sites. The quasienergy convention is $U|\phi\rangle = e^{-i\theta}|\phi\rangle$, $\theta \in (-\pi, \pi]$. The bipartition is $A =$ sites $1 \ldots \lfloor N/2 \rfloor$.

4.2 Pipeline and independent cross-checks

The diagonalisation is done in Julia by `HSF/generate_spectral_data.jl`, an optimised successor of the earlier `generate_EE_data.jl`, which is left untouched. The analysis is done in Python, in `code/qca_fragmentation/spectral/`.

- **Two independent constructions of U** agree. The Julia operator is built from Kronecker products of local gates. The Python operator comes from this repository's exact $\mathbb{Z}[1/\sqrt{2}]$ engine (`core/cycle.py`). Their quasienergy multisets agree to $< 10^{-14}$ for all 16

rules at $N = 8$ (test `test_engine_matches_julia_spectrum`, $N = 6, 7, 8$). Comparisons of angle multisets cut the circle in the middle of the largest gap. Sorting raw angles is unsafe because a level at $\theta = 0$ computed as -10^{-16} wraps to 2π. A first version of the test fell into exactly this trap and reported a spurious disagreement.

- **Against the stored HSF eigendata** (complex `eigen` per sector): quasienergies agree to 10^{-14}. The entropies agree to 10^{-15} for every globally non-degenerate level, the only case where the old entropies are basis independent.

4.3 Diagonalisation (details and optimisation in Sec. 5)

U is real orthogonal (Sec. 3). Each block is diagonalised by

1. solving the real symmetric problem $C = (A + A^T)/2$, whose eigenvalues are $\cos\theta$, with LAPACK's MRRR solver ($O(n)$ workspace);
2. grouping the $\cos\theta$ into clusters (tolerance 10^{-9}) and diagonalising the antisymmetric part $(A - A^T)/2$ inside each cluster, which separates θ from $-\theta$.

This yields exactly orthonormal eigenvectors, also inside degenerate eigenspaces, which a general complex eigensolver does not guarantee. It is also several times faster. The three dense products the cluster loop needs are done as single matrix–matrix multiplications rather than per cluster (that step was memory-bound and took half the run time). On the 12th-gen i7 used here, dense LAPACK is fastest with 4–6 BLAS threads and degrades sharply at 20. Every run stores the largest eigen-residual $\max |Uv - e^{-i\theta}v|$ and the orthonormality error.

4.4 Symmetry resolution

Each eigenvector carries three labels, all found and verified at run time:

1. **Krylov sector**: connected component of the support of U (classical fragmentation).
2. **Gate-algebra block**: the decomposition of each sector by the commutant of the bond algebra $\mathcal{A}_0 = \langle g_1, \ldots, g_N \rangle$ generated by the local gates [28]. This includes quantum fragmentation and hidden internal symmetries. It is found without prior knowledge of the symmetry:
   (a) diagonalise a generic element $K = \sum_i a_i g_i$ (random $a_i \in [1, 2]$, fixed seed);
   (b) transform a second generic element $K' = \sum_i b_i g_i$ into that eigenbasis;
   (c) take the connected components of the transformed K' as the blocks.

   If the commutant is Abelian, each eigenvector of K lies in one block, and K' connects everything within a block. Isomorphic blocks (non-Abelian multiplicity) are merged, which is harmless: they give exactly degenerate copies of one spectrum, removed in the statistics. *Check*: the largest matrix element of $Q^T U Q$ between different blocks is stored (`max_offblock`).
3. **Spatial label.**
   - obc0, odd N, mirror-symmetric rules ($r_{01} = r_{10}$): the reflection R commutes with U, and every eigenvector gets a parity ± 1, a true quantum number.
   - obc0, even N: R exchanges the two layers ($RL_AR = L_B$), so R alone is no symmetry. The half step $\Omega = RL_A$ is one: it commutes with U and $\Omega^2 = U$ (both asserted at run time).

- pbc (even N only), every rule: translation by *one* site maps layer A onto layer B, so the half-translation $\Omega_T = T_1 L_A$ commutes with U and $\Omega_T^2 = UT_2$, with T_2 the translation by two sites (asserted). It resolves the T_2 momentum $j = 0, \ldots, N/2-1$ and splits each momentum sector once more. The site-centred reflection does not commute with Ω_T in general and is not resolved on the ring: a residual $\mathbb{Z}_2$ can remain in the $j = 0$ and $j = N/4$ sectors of mirror-symmetric rules.

For the square-root symmetries (Ω, Ω_T) the stored quantity is their eigenphase φ, obtained by diagonalising the symmetry inside each degenerate eigenspace of U (with the same orthonormal cos/sin two-step as for U). The resolved spectrum of a block is the spectrum of φ, not of θ: a ± 1 "sheet" label derived from $e^{i\theta/2}$ would flip at the branch cut $\theta = \pm\pi$ and mix the two halves there. Labels are used for a block only if the block is checked to be invariant under the symmetry, and the invariance residual is stored.

Time reversal ($T = L_A K$, $T^2 = 1$) fixes the reference ensemble to $\beta = 1$ (Sec. 3).

4.5 Statistics

Gap ratios are computed per fully resolved block (sector, gate-algebra block, true symmetry label), on the block's level variable (φ where a square-root symmetry was resolved, otherwise θ), with exact degeneracies removed.

- *Folding is decided numerically per block.* If the block spectrum is symmetric under $x \to -x$ (complex conjugation maps the block to itself: every block of a real symmetry), only $|x| \in [0, \pi]$ is used. Otherwise, e.g. in a ring-momentum sector $j \neq -j$ whose conjugate partner is another sector, the full circle is used, including the wrap-around spacing.

- *The reference ensemble depends on the block.* Self-conjugate blocks carry the time reversal $T = L_A K$ and are compared with $\beta = 1$ (0.5307). In momentum sectors $j \neq -j$, T maps $j \to -j$, so no antiunitary symmetry is left within the sector, and a chaotic sector is $\beta = 2$ (CUE, 0.5996). Both branches are tested against random matrices: CUE through the full-circle branch gives 0.597 ± 0.005, and products of random reflections through the folding branch give 0.534 ± 0.007.

- Only blocks with at least 18 ratios are used, and the ratios of all such blocks are pooled. For comparison the same statistic is computed at two coarser resolutions: per Krylov sector only, and for the whole spectrum as one list.

4.6 Entanglement and the fair reference

For every eigenvector the half-chain von Neumann entropy S_{vN} is stored. For every degenerate eigenspace of dimension g (inside its block and label) the basis-independent Haar-averaged Rényi-2 entropy is stored as well,

$$S_2^{\mathrm{Haar}} = -\ln \frac{\operatorname{Tr} P_A^2 + \operatorname{Tr} P_B^2}{g(g+1)}, \qquad P_A = \operatorname{Tr}_B P,\ P_B = \operatorname{Tr}_A P, \tag{2}$$

where P is the projector onto the eigenspace. This follows from $\mathbb{E}[|\psi\rangle\langle\psi|^{\otimes 2}] = P^{\otimes 2}(1 + \mathrm{SWAP})/(g(g+1))$, and for $g = 1$ it is the ordinary Rényi-2 entropy of the eigenvector. The same formula with P the projector onto the whole *block* gives $S_2^{\mathrm{rand}}(\mathrm{block})$, the entanglement of a random state inside that block. That is the fair reference for a fragmented system (Sec. 3), and the ratio

$$\eta = S_2^{\mathrm{Haar}} / S_2^{\mathrm{rand}}(\mathrm{block}) \tag{3}$$

is ≈ 1 for a thermal eigenstate and $\ll 1$ for an atypically weakly entangled, scar-like one. η is defined for blocks of dimension ≥ 2. States with $\eta < 0.5$ in blocks of dimension ≥ 20 are counted as scar-like.

5 Algorithm and optimisation choices

This section documents the production code, `HSF/generate_spectral_data_v2.jl` (Julia), step by step. For every design choice it gives the reason, the measured effect, and how the change was validated. The original pipeline (`generate_EE_data.jl`, untouched) and the first rewrite (v1, `generate_spectral_data.jl`) serve as references. All timings are on one 12th-generation Intel i7-12700H (6 performance + 8 efficiency cores, 20 hardware threads, 31 GB), with OpenBLAS.

5.1 The algorithm

For one unit (rule r, size N, gate $V \in \{H, X\}$, boundary condition obc0 or pbc):

1. **Local gates from bit operations.** Each controlled gate g_i is a sparse $2^N \times 2^N$ matrix with at most two entries per column. It is built by reading the two neighbour bits of each basis state (vacuum outside the chain for obc0, wrap-around for pbc). For obc0 every g_i is asserted equal to HSF's Kronecker-product construction.

2. **The Floquet operator is never formed.** $L_A = \prod_{i \text{ odd}} g_i$ and $L_B = \prod_{i \text{ even}} g_i$ are sparse (at most $2^{N/2}$ entries per column). U is applied as $Uv = L_B(L_A v)$.

3. **Krylov sectors.** For $V = H$: union-find over the supports of the individual gates. This is the same partition as the connected components of $\mathrm{supp}(U)$ (R-T19 Prop. 4), asserted at run time for every $N \leq 10$. For $V = X$, U is a permutation and is formed directly; its orbits are the sectors.

4. **Spatial symmetry**, detected and verified by random-vector tests through the layers:

   - obc0: the reflection R if $RUR = U$; otherwise the half step $\Omega = RL_A$, if $RL_AR = L_B$ (asserting $\Omega^2 = U$ and $[\Omega, U] = 0$).
   - pbc: translation T_2 by two sites, and the site-centred reflection if it commutes with U.

5. **Per Krylov sector** S ($d = |S|$):

   (a) *Gate-algebra blocks* (Sec. 4.4): diagonalise $K = \sum_i a_i g_i|_S$ (random a_i, fixed seed), transform $K' = \sum_i b_i g_i|_S$ into that eigenbasis Q, and take connected components. Skipped for $V = X$, whose orbits are finer than the gate-algebra blocks, and for $d \leq 2$.

   (b) *Dense sector block* U_S, assembled from the layers in column chunks. Then $Q^T U_S Q$, and each block $A_b = Q_b^T U_S Q_b$. Check: the largest off-block element is stored.

   (c) *Per block*: real-symmetric eigensolve of $C = (A_b + A_b^T)/2$; clustering of $\cos\theta$; within each cluster, the Hermitian matrix $iV_c^T S_b V_c$, with $S_b = (A_b - A_b^T)/2$, separates θ from $-\theta$. Then quasienergy groups (degeneracies); symmetry labels by successive diagonalisation inside each degenerate eigenspace; entanglement.

6. **Output** (JLD2, readable with h5py): per eigenvector the sector, block, symmetry label, θ, S_{vN}, degeneracy, S_2^{Haar} of its eigenspace, and S_2^{rand} of its block. Per unit: the sector and block sizes, the symmetry, and five checks (eigen-residual, orthonormality, off-block leakage, symmetry residual, run time).

5.2 Optimisation choices, with measured effect

1. Real-symmetric eigensolver instead of a complex non-Hermitian one. U is real orthogonal, so $C = (U + U^T)/2$ is real symmetric with eigenvalues $\cos\theta$, and it commutes with $S = (U - U^T)/2$.

- *Speed:* at $n = 2048$ (4 BLAS threads) the complex `eigen(U)` takes 11.0 s, a real non-symmetric `eigen(U)` 4.4 s, and the real-symmetric `eigen!(Symmetric(C))` 2.2 s, a factor 5.
- *Correctness:* a general eigensolver does not return orthonormal eigenvectors inside degenerate eigenspaces, and these spectra are highly degenerate (up to 95 % of the states for 105/150). The symmetric solver does, so every stored entanglement belongs to a properly normalised, mutually orthogonal basis.
- *Splitting:* θ and $-\theta$ share $\cos\theta$. Inside each cos-cluster, S restricted to the cluster is antisymmetric, and iS is Hermitian with eigenvalues $\pm\sin\theta$. Clusters are tiny (mostly pairs), so this costs nothing.
- *Workspace:* the MRRR driver needs $O(n)$ workspace, against $O(n^2)$ for divide and conquer, which matters at $n = 16384$.

2. BLAS threading. Dense LAPACK on this hybrid CPU is fastest with 4–6 BLAS threads. Measured for a 4096×4096 symmetric eigensolve:

BLAS threads	1	4	10	20
eigensolve (s, Julia −t 1)	9.7	5.1	6.4	32.9
eigensolve (s, Julia −t 20)	11.6	5.7	6.6	7.0

Production uses 6 BLAS threads for the dense stages and Julia's 20 threads (with BLAS set to 1) for the per-cluster loop.

3. Batched products instead of per-cluster ones. The first version multiplied the full $d \times d$ matrices by each cluster's few columns separately. With about $d/2$ clusters, this streams the full matrices from memory $d/2$ times and is memory-bound: 39 of 74 s at $N = 12$ (rule 147). Doing AV, SV and EV as matrix–matrix products cut the run to 35 s.

4. Never forming U (the memory incident). Profiling the phases at $N = 12$ showed 1.1 GB live after *building U and finding its sectors*, before any dense algebra. For the chaotic rules U is effectively dense: a sparse matrix with up to 2^N entries per column, plus Graphs.jl's adjacency lists for the component search. That extrapolates to about 18 GB at $N = 14$, which is what triggered the memory reaper. Applying U through the two sparse layers, and finding sectors from the individual gates, removes this entirely. Peak memory for rule 147 at $N = 13$: 7.4 GB (v1) $\rightarrow$ 4.5 GB (v2); run time 224 s $\rightarrow$ 124 s.

5. Shared lookup instead of sparse slicing. `SparseArrays`' `getindex` allocates an m-length cache per call. With N gates sliced per sector and 10^3–10^4 sectors at $N \geq 16$, this alone would cost terabytes of memory traffic. All restrictions now use one shared position vector, set and reset per sector.

6. Entanglement in compressed coordinates. The entanglement of an eigenvector supported on a sector S only involves the rows and columns of the $2^{N_B} \times 2^{N_A}$ reshaped matrix that S occupies. Zero rows and columns change neither the singular values nor $\operatorname{Tr} P_A^2$ and $\operatorname{Tr} P_B^2$, so the code works with the occupied rows and columns only. For fragmented rules this shrinks every SVD from 256×256 (at $N = 16$) to a few dozen. Effect for rule 156: $N = 16$ from > 10 minutes (aborted) to 15 s; $N = 18$ in 36 s with 1.3 GB peak. These are what make the per-rule larger sizes possible.

7. Smaller choices.

- The orthonormality check uses 64 sampled columns for blocks larger than 2048 (the full V^TV would be another $d \times d$ matrix).
- Garbage collection is forced only after blocks with $d > 2000$. Forcing it after every block made rule 156 at $N = 9$ ten times slower (2.2 s $\rightarrow$ 21 s), since it has hundreds of tiny blocks.
- Block bases are stored as contiguous copies, not views. A view with non-contiguous columns made EV fall back from BLAS to Julia's generic single-threaded product (observed: about 1.2 cores busy).
- Q^TXQ is formed in column chunks. This did *not* lower the measured peak (4.5 GB either way): the peak sits in the block eigensolve itself (block, basis, eigenvectors, LAPACK workspace).
- Julia runs with `--heap-size-hint=14G`, so the collector works harder before the machine runs short.

Expected peak at $N = 14$ (a 16384-dimensional block): about 15 GB.

5.3 How every change was validated

No optimisation was accepted on the strength of its own checks. Each one was compared against the previous version, and the whole chain against independent constructions:

comparison	scope	result
v1 vs HSF eigendata (complex `eigen` per sector)	rules 4, 6, 15, $N = 9$	θ to 10^{-14}; S_{vN} (non-degenerate) to 10^{-15}
exact Python engine vs Julia	all 16 rules, $N = 6, 7, 8$	spectra to $< 10^{-9}$ (test suite)
v2 vs v1, after each change	6–8 rules, $N = 9$	block by block identical[†]
v2 (pbc) vs exact engine (pbc)	7 rules, $N = 8$	spectra to $< 10^{-14}$
run-time assertions	every run	gates = kron (obc0); sectors = components of supp(U) ($N \leq 10$); $\Omega^2 = U$, $[\Omega, U] = 0$, $[T_2, U] = 0$

[†] Compared as multisets per block: sector size, block size, sorted quasienergies, S_2^{Haar}, degeneracies, S_{vN} of the non-degenerate states, and S_2^{rand}.

5.4 Pitfalls caught, recorded so they are not repeated

1. **Shared variables in threaded closures (Julia).** Twice, a variable assigned inside a `Threads.@threads` loop had the same name as one in the enclosing function (`SV`, then `G`). Julia then shares one boxed variable across threads, a race that produced dimension errors. Fixed by renaming. Any new variable in `process_block!` must not reuse a name assigned in the loop body.
2. **Comparing angles on a circle.** Sorting raw angles misaligns lists when a level at $\theta = 0$ is computed as -10^{-16} and wraps to 2π. A first comparison therefore reported a spurious 0.7 rad disagreement and briefly stopped production. All comparisons now cut the circle in the largest gap (`floquet.circle_distance`).
3. **Sort-key comparisons.** Pairing states by a sort on (sector, block, θ rounded) misaligns nearly degenerate levels and reported a false mismatch between two correct versions. All comparisons are now block-wise multisets.

4. **Shell tooling.** `pkill -f` with a pattern that also appears in one's own command line kills that command. A `for` loop over paths split them at the spaces in "QCA Simulations" (caught after one minute; the deletions targeted non-existent fragments, so no file was lost).

5. **An incomplete symmetry analysis on the ring.** The first pbc runs resolved only T_2 and folded every block to $|\theta|$. The chaotic rules then came out at $\langle r\rangle \approx 0.43$ on the ring against 0.53 with open boundaries, a typical two-sector mixture. Two errors were responsible. First, the half-translation $\Omega_T = T_1 L_A$ commutes with U for every rule on an even ring and was not resolved. Second, a momentum sector $j \neq -j$ is not conjugation-symmetric, so folding it pairs unrelated levels. Both were fixed (Sec. 4.4), the 64 affected files regenerated (the old ones kept in `results_spectra/H_pbc_T2only/`), and the statistics given a numerical fold test. The same analysis showed that the ± 1 label of Ω (obc0, even N) flips at the branch cut. Its blocks were therefore regenerated with the eigenphase φ stored.

6. **The memory reaper.** Background jobs that push the machine to critically low memory are stopped by the session and must not be restarted without permission. This was the motivation for items 4–7.

6 Results: the sixteen rules with the Hadamard gate

6.1 The organising principle: how many neighbour patterns fire the gate

Count, for each rule, the number k of the four neighbour patterns $(x_{i-1}, x_{i+1}) \in \{00, 01, 10, 11\}$ on which the Hadamard acts. For $k = 2$, also record whether the firing set depends on only one neighbour ("one-sided": $60, 102, 153, 195$) or on both ("XOR": 150 fires on $01, 10$, and 105 on $00, 11$). This single combinatorial number sorts the 16 rules into classes with distinct spectral signatures (Table 2, Fig. 1):

class	rules	spectral signature (obc0)
$k = 0$	204	$U = I$ (identity)
$k = 1$	$108, 156, 198, 201$	exponential fragmentation; Poisson inside the pooled blocks
$k = 2$ one-sided	$60, 102, 153, 195$	a tower of gate-algebra blocks; $\langle r\rangle$ near Poisson, drifting up with N. *On a ring: no tower, chaotic* (Sec. 7)
$k = 2$ XOR	$105, 150$	arithmetic spectrum, 3^m distinct quasienergies (W150)
$k = 3$	$54, 57, 99, 147$	**chaotic (GOE)**, plus a fixed number of exact gate-singlet eigenstates. On a ring: chaotic, CUE or COE by symmetry
$k = 4$	51	$U = H^{\otimes N}$, non-interacting

6.2 $k = 3$: chaos, with exact scars that do not grow

With open boundaries, the four rules on which the gate fires on all but one pattern are the only chaotic ones. (On a ring, the one-sided $k = 2$ rules join them; Sec. 7.)

- **57 and 99** (mirror images; computed independently, identical to all digits): $\langle r\rangle = 0.5311 \pm 0.0028$ at $N = 14$ (8190 ratios), on the GOE value 0.5307 from $N = 10$ on: $0.525, 0.523, 0.527, 0.535, 0.531$ for $N = 10 \ldots 14$. $P(r)$ follows the $\beta = 1$ surmise over the whole range (Fig. 2). The eigenstates in the big block are thermal *within that block*: $\bar{\eta} = 0.95$.

k	W	N	sym.	sectors	blocks	$\langle r\rangle_{\text{none}}$	$\langle r\rangle_{\text{full}}$	$\#r$	distinct $\theta/2^N$	$\bar{\eta}_{\text{big}}$	singlets
0	204	16	R	65536	65536	–	–	0	1.53e-05	–	0
1	108	19	R	55405	124345	0.386	0.392	71048	0.0303	0.55	52688
1	156	19	none	10946	121370	0.412	0.421	5486	0.00408	0.36	37812
1	198	19	none	10946	121370	0.412	0.421	5486	0.00408	0.35	37812
1	201	19	R	31572	90053	0.383	0.389	97266	0.068	0.54	39872
2 one-sided	60	14	none	15	105	0.384	0.395	7732	0.492	0.68	24
2 one-sided	102	14	none	15	105	0.384	0.395	7732	0.492	0.68	24
2 one-sided	153	14	none	1	15	0.398	0.421	8092	0.992	0.81	2
2 one-sided	195	14	none	1	15	0.398	0.421	8092	0.992	0.81	2
2 XOR	105	15	R	9	17	0.491	0.498	1512	0.0186	0.81	7
2 XOR	150	15	R	9	23	0.404	0.389	5987	0.1	0.66	7
3	54	14	Omega	2	3	0.428	0.523	8127	0.992	0.84	1
3	57	14	none	1	3	0.531	0.531	8190	1	0.96	2
3	99	14	none	1	3	0.531	0.531	8190	1	0.96	2
3	147	14	Omega	1	5	0.422	0.527	8117	0.991	0.89	4
4	51	14	R	1	16384	–	–	0	0.000122	–	16384

Table 2: Per-rule summary at each rule's largest computed N (obc0, gate H), generated by `spectral.analysis.write_table`. sym.: spatial symmetry resolved (R for odd N; the half-step Ω for even N). sectors/blocks: Krylov sectors, and fully resolved gate-algebra blocks. $\langle r\rangle_{\text{none}}$: whole spectrum as one list. $\langle r\rangle_{\text{full}}$: pooled over fully resolved blocks with ≥ 20 distinct half-levels, from $\#r$ ratios. distinct $\theta/2^N$: number of distinct quasienergies over the whole spectrum. $\bar{\eta}_{\text{big}}$: mean S_2/S_2^{rand}(block) in the largest block. singlets: one-dimensional gate-algebra blocks inside Krylov sectors of dimension ≥ 3, i.e. exact eigenstates of every individual gate.

- **147**: $\langle r\rangle_{\text{full}} = 0.526$ ($N = 13$) and 0.5274 ± 0.0028 ($N = 14$), but only after resolving the reflection (odd N) or the half step Ω (even N). Unresolved, the same spectra give 0.424 and 0.422. This is the textbook symmetry-mixing effect, and here it is the difference between "integrable" and "chaotic".
- **54**: $\langle r\rangle_{\text{full}}$ rises steadily, $0.440, 0.456, 0.502, 0.510, 0.523$ for $N = 10 \ldots 14$, towards GOE. Its frozen vacuum $|0 \ldots 0\rangle$ is a separate Krylov sector.

Gate singlets. Inside the chaotic block, each of these rules carries one-dimensional blocks of the gate algebra: states that are eigenvectors of *every individual gate* g_i, hence of U for any order and any number of gate applications. Their number does not depend on N ($N = 6 \ldots 14$):

- two for 57 and 99, and four for 147, all *product* states ($S = 0$) at $\theta \in \{0, \pi\}$;
- exactly one for 54, at $\theta = 0$, which is *entangled*. Its entanglement *decreases* with N: $S_{\text{vN}} = 0.470, 0.430, 0.405, 0.371, 0.345, 0.315, 0.291, 0.265, 0.243$ for $N = 6 \ldots 14$ (S_2 from 0.35 to 0.13). It is an area-law state that approaches a product state as the chain grows. (An earlier draft of this report said "0.29 at every N"; that generalised from the single value at $N = 12$ and is withdrawn.)

All of them descend from one universal eigenstate, $|h_+\rangle^{\otimes N}$ (Sec. 10). The entangled singlet of 54 is its projection off the frozen vacuum, $\propto |h_+\rangle^{\otimes N} - \cos^N(\pi/8)|0 \ldots 0\rangle$, whose entropy follows in closed form and reproduces the list above. The $\theta = \pi$ states put $|h_-\rangle$ on a boundary site whose gate fires whatever its inner neighbour is. These are exact many-body scars in the sense of Refs. [20, 23]: non-thermal eigenstates embedded in a GOE spectrum. The embedding is exact here, by the gate algebra, not approximate. Because they are eigenstates of every gate, they are also stationary under any reordering of the circuit, which a Hamiltonian scar cannot offer.

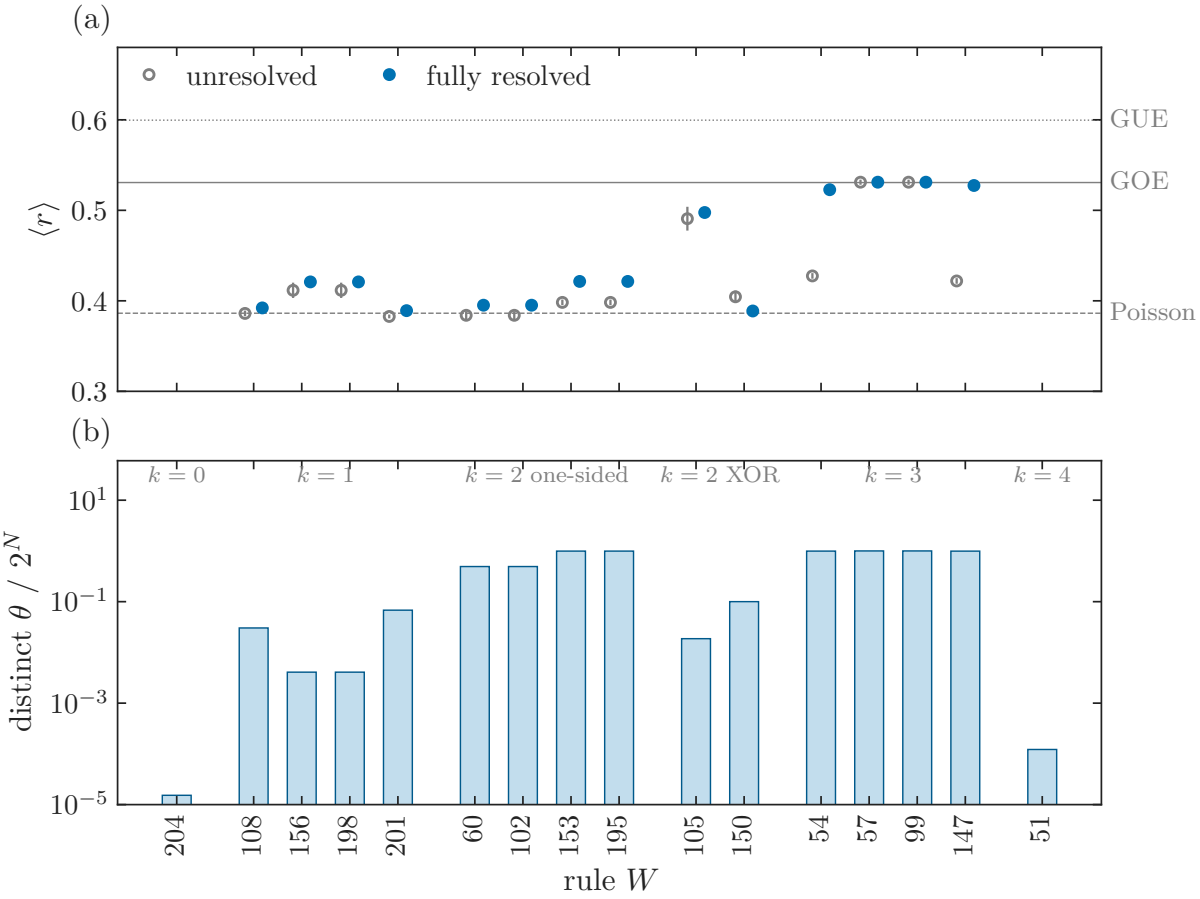


Figure 1: (a) Gap ratio of every rule at its largest N: unresolved (whole spectrum, open circles) and fully resolved (Sec. 4.4, filled). Lines: Poisson, GOE ($\beta = 1$, the class of U), GUE. (b) Fraction of distinct quasienergies in the whole spectrum (log scale). Rules grouped by the number k of firing neighbour patterns.

N-stable outliers in 57/99. Besides the singlets, 57 has a handful of anomalously weakly entangled eigenstates in the big block ($\eta \approx 0.2$): 4 at $N = 11$ and 13, and 10 at $N = 12$. Their quasienergies are stable with N ($\theta/\pi = \pm 0.6877 \to \pm 0.6879$ from $N = 11$ to 13, and ± 0.3122 at $N = 12$ and 13). They do not form an equally spaced tower, and their number does not grow. They are an edge mode bound to the left end on top of the $|h_+\rangle^{\otimes N}$ vacuum. The two quasienergies are one mode, $\pm 0.3122\pi$, the second shifted by π by the right-end $|h_-\rangle$ singlet (Sec. 10). They are not a PXP-type scar band. In 147, by contrast, the low-η states are numerous (72–83) and spread over θ: the finite-size tail of the thermal distribution.

6.3 $k = 1$: exponential fragmentation; two pairs at the same N

Rules 108, 156, 198 and 201 fire on a single pattern (108 on 11, 201 on 00, 156 on 10, 198 on 01) and have exponentially many Krylov sectors (R2, R19). They were run to $N = 20$ (201 to $N = 19$; see below). The class is compared at the largest N that all four rules share, $N = 19$ (Table 2), and not each rule at its own maximum.

- **108 and 201** (fire on 11 or 00; mirror symmetric) have large blocks: the largest has 2583 and 5544 states at $N = 19$. Pooled over them, $\langle r \rangle = 0.3922 \pm 0.0010$ (108) and 0.3891 ± 0.0009 (201), with 7×10^4 to 10^5 ratios. This is Poisson (0.3863) up to a small, statistically resolved excess of 0.003 to 0.006, which is stable from $N = 17$ on.

- **156 and 198** (fire on 10 or 01; a reflection pair, computed independently and identical) stay *tiny*: the largest block has 100 states at $N = 19$ (125 at $N = 20$), and only 2139 of the 2^{19} quasienergies are distinct. $\langle r \rangle = 0.4209 \pm 0.0038$ at $N = 19$ and 0.4172 ± 0.0023 at $N = 20$, stable from $N = 17$ on. This is clearly *above* Poisson but far below GOE.

The block structure explains both values. The commutant study (R-T18/R-T27) shows that a block is a tensor product of independent rooms, so its quasienergies are sums of room quasienergies. When many rooms of different sizes contribute (108, 201), the sums are effectively independent and the statistics are Poisson. When a block is built from a few small, repeated rooms (156, 198), the sums are *structured*. This is now exact (Sec. 10): each segment between frozen 01 walls holds one mobile kink with $\cos\theta_m = -\sin^2(\pi m/(L+1))$, and a block spectrum is the set of sums over its short segments, so a handful of values recur with many exact and near coincidences. Neither pair shows level repulsion growing with N.

The gate singlets are exponentially many (at $N = 19$: 52 688 for 108, 37 812 for 156 and 198, and 39 872 for 201; roughly doubling with each added site). These are the dark states of the rooms.

6.4 $k = 2$, XOR: arithmetic spectra

150 and 105 have only a handful of Krylov sectors (the domain-wall shells, R8) and large blocks, but their spectra are *arithmetic*. The number of distinct quasienergies of 150 is exactly 3^m for even N and $(3^m + (-1)^m)/2$ for odd N ($m = \lceil N/2 \rceil$) at every $N = 6 \ldots 15$, reproducing R8's law from an independent diagonalisation. 105 shares the even-N counts but has far fewer at odd N (251 against 1093 at $N = 13$). Up to 95 % of the eigenstates lie in degenerate eigenspaces, and spacings repeat exactly. $\langle r \rangle$ (0.39 for 150 and 0.50 for 105 at their common $N = 15$) is therefore no measure of chaos here. The right description is the free wall walk with sign defects of R8. Equivalently, 150 with $V = H$ is a free-fermion circuit [40], which explains both the degeneracies and the exact level counting. Each Krylov sector contains one gate singlet at $\theta = 0$, an entangled dark state (S between 0.13 and 1.14).

6.5 $k = 2$, one-sided: a boundary tower with intermediate statistics

In 60/102 (gate fires iff the left or right neighbour is 1) and 153/195 (iff it is 0), the gate on a site is controlled by one neighbour only. With open boundaries the gate algebra then produces a tower:

- 153/195: $N + 1$ blocks of sizes $2^{N-1}, 2^{N-2}, \ldots, 2, 1$, plus two singlets;
- 60/102: $2N - 4$ singlets and many small blocks.

The tower is exact. For 153 the projectors $Q_j = \prod_{s<j} P_s^+ P_j^-$ onto Hadamard eigenstates commute with U, so block j is $|h_+\rangle^{\otimes j}|h_-\rangle\otimes$(the remaining sites), i.e. the same chain shortened, with one boundary site fixed. For 60 the same holds from the right end, inside the sector "site 0 = 1" (Sec. 10). The entanglement accordingly falls into bands, one per block (Fig. 4). Within the blocks, $\langle r \rangle$ is near Poisson but *increases* steadily with N: for 60, $0.334, 0.351, 0.360, 0.369, 0.381, 0.395, 0.405$ for $N = 9 \ldots 15$; for 153, 0.370 to 0.421 for $N = 9 \ldots 14$. Built directly from the tower (Sec. 10), the largest block reaches $N = 15$ for 153 and $N = 16$ for 60. There the drift stops: $\langle r \rangle = 0.4236$ (60, $N = 14$), 0.4216 ($N = 15$) and 0.4238 ($N = 16$), and 0.4239 and 0.4236 for 153 at $N = 13$ and 15. On the open chain the statistics plateau at an intermediate value; they are not heading to GOE. The tower itself is a boundary effect. On a ring it is absent, and the same rules are chaotic with $\beta = 1$ statistics ($\langle r \rangle = 0.527$ and 0.523 for 153 and 60 at $N = 16$; Sec. 7), protected by a dressed-reflection time reversal (Sec. 10). So on the open chain, the non-ergodicity of this class is produced by the ends.

6.6 $k = 0$ and $k = 4$: trivial

204 is the identity (one distinct quasienergy). In 51 every site gets H in its layer, so $U = H^{\otimes N}$ exactly. Its spectrum is $\{0, \pi\}$, every state is a gate singlet, and the eigenstates are products of single-site H eigenstates, with $S = 0$.

7 Periodic boundary conditions

The user asked whether the classes survive a change of boundary condition. The obvious suspects are the one-sided rules, whose block tower is anchored at the open ends. All sixteen rules were run on rings of even length $N = 6 \ldots 14$. At $N = 14$, the eight rules with $k \geq 2$ that are not arithmetic have one $\approx 16\,000$-dimensional sector each, and each run took about 16 min.

7.1 What is resolved on the ring, and which ensemble to expect

On an even ring, translation by one site T_1 maps layer A onto layer B, for every rule. The half-step

$$\Omega_T = T_1 L_A, \qquad [\Omega_T, U] = 0, \qquad \Omega_T^2 = U\,T_2, \tag{4}$$

is therefore a symmetry of every rule. Its eigenphase φ fixes both the quasienergy and the momentum j of the two-site translation T_2 ($e^{2i\varphi} = e^{-i\theta} e^{2\pi i j/(N/2)}$). A statistics block is (Krylov sector, gate-algebra block, j), and its level variable is φ. The site reflection R does *not* commute with Ω_T and is not resolved. Which random-matrix ensemble applies depends on the type of sector:

- *Real sectors* ($j = 0$, and $j = N/4$ when $N/4$ is an integer) are mapped to themselves by complex conjugation K. At $j = 0$, T_1 is an involution, so Ω_T is again a product of two reflections: $\beta = 1$ on the half spectrum (rule 57: 0.549 at $N = 14$). At $j = N/4$, $T_1^2 = -1$ and the reflection structure is lost, so the reference is a real orthogonal matrix without that structure, which has $\beta = 2$-like local statistics (rule 57: 0.610 at $N = 12$). For mirror-symmetric rules, the real sectors also contain the unresolved $\mathbb{Z}_2$ generated by $L_A R$, which pushes $\langle r \rangle$ down. **The real sectors are therefore not used as the diagnostic.**

- *Complex sectors* ($j \neq -j$) carry most of the levels, so they are the diagnostic. K maps $j \to -j$, so a time-reversal symmetry inside sector j needs a unitary P that also flips j and satisfies $PUP^{-1} = U^{-1}$. For a mirror-symmetric rule the site reflection does this (R exchanges the layers), so the naive reference is COE ($\beta = 1$, 0.5307). For a rule without mirror symmetry the naive reference is CUE ($\beta = 2$, 0.5996).

The block classification (real or complex) is numerical: a block counts as real if its eigenphases are closed under conjugation to 10^{-7}.

Correction during the analysis. The first version of this conjugation test compared the sorted lists of x and $-x$ element by element. A level at $0 \equiv 2\pi$ (the $j = 0$ sector has forced eigenvalues ± 1) moves to the other end of the sorted list and shifts every pair. The test then wrongly classified $j = 0$ as complex, so its full circle was used. The fix (`stats.conj_symmetric`) matches each $-x$ to its nearest level on the circle. A test pins the $j = 0$ block of rule 57 at $N = 8$ against the exact Python engine. The obc0 numbers of Sec. 6 were recomputed: none of the 145 units changed.

7.2 Results

$k = 3$: the class survives, and the ensemble follows the symmetry. 57 and 99 have no mirror symmetry and become CUE on the ring: $\langle r \rangle = 0.602 \pm 0.002$, with a GUE-shaped $P(r)$

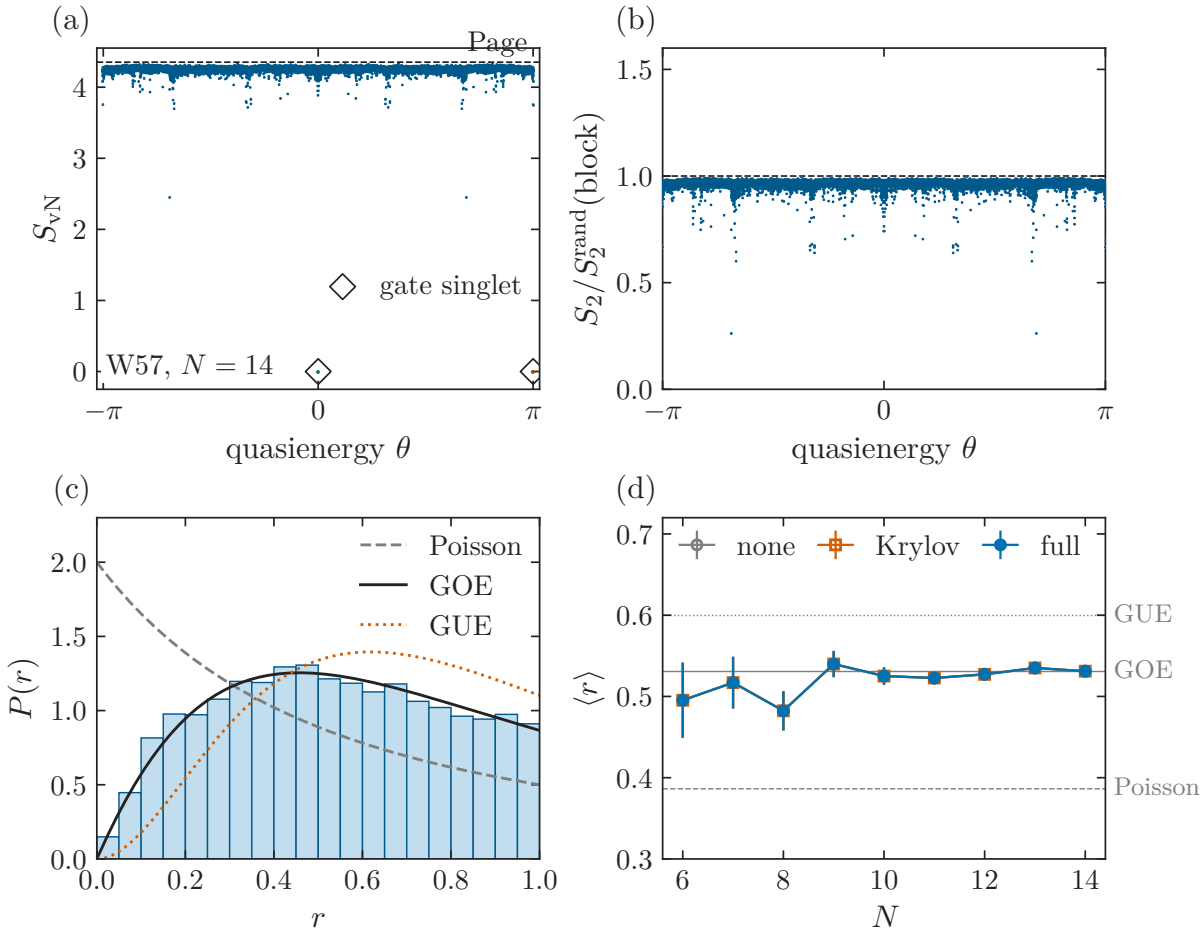


Figure 2: Rule 57 ($k = 3$), largest N, obc0. (a) Half-chain entanglement against quasienergy. Coloured: non-degenerate states, by block. Grey: states in a degenerate eigenspace, whose entanglement depends on the basis. Diamonds: gate singlets (exact eigenstates of every gate; product states at $\theta = 0, \pi$). (b) $\eta = S_2^{\mathrm{Haar}}/S_2^{\mathrm{rand}}(\mathrm{block})$, basis independent. (c) $P(r)$ of the fully resolved half spectrum against the Poisson, GOE and GUE surmises. (d) $\langle r \rangle$ against N at three resolution levels.

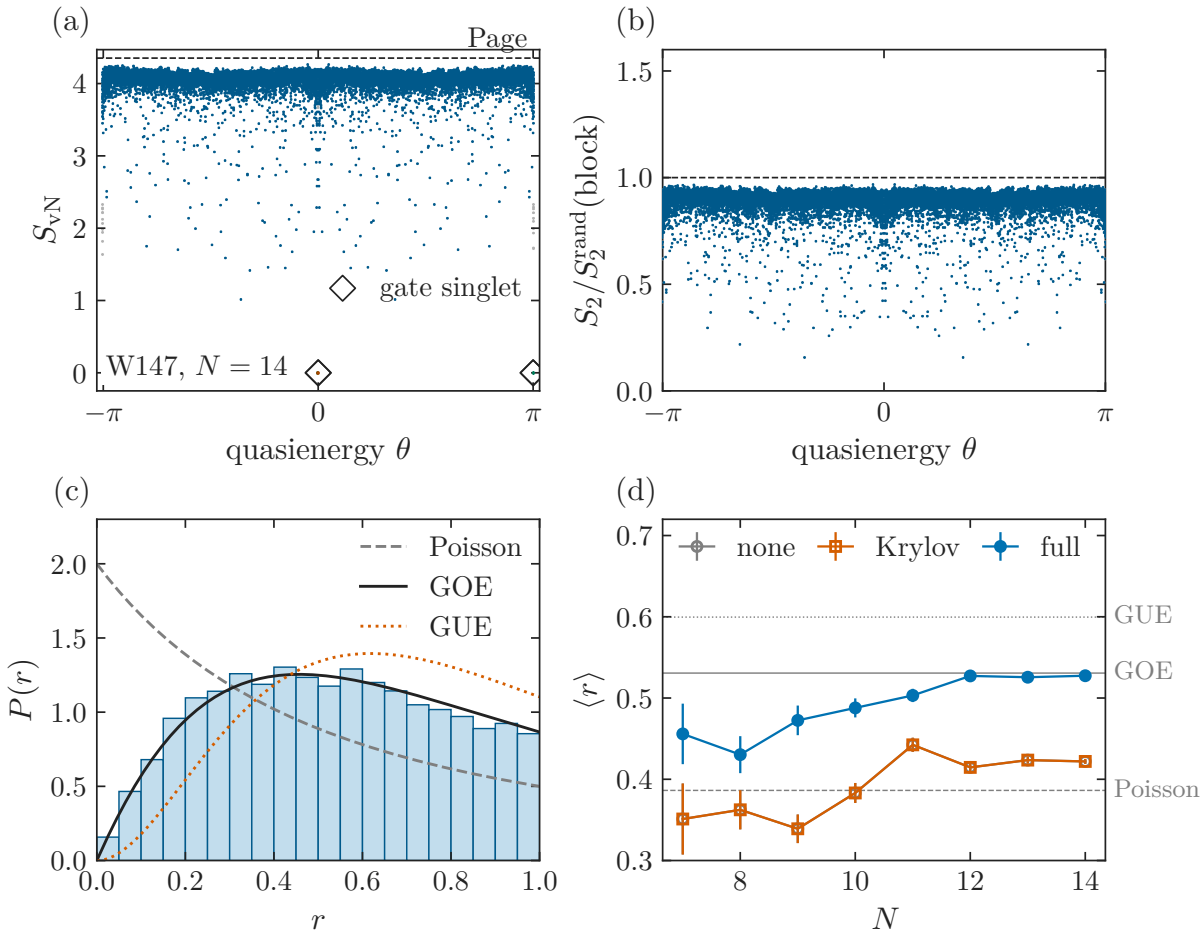


Figure 3: Rule 147 ($k = 3$), largest N; panels as in Fig. 2. In (d) the Krylov-only statistics (here the whole spectrum) stay near Poisson, while resolving the spatial symmetry (R for odd N, the half step Ω for even N) brings $\langle r \rangle$ to the GOE value.

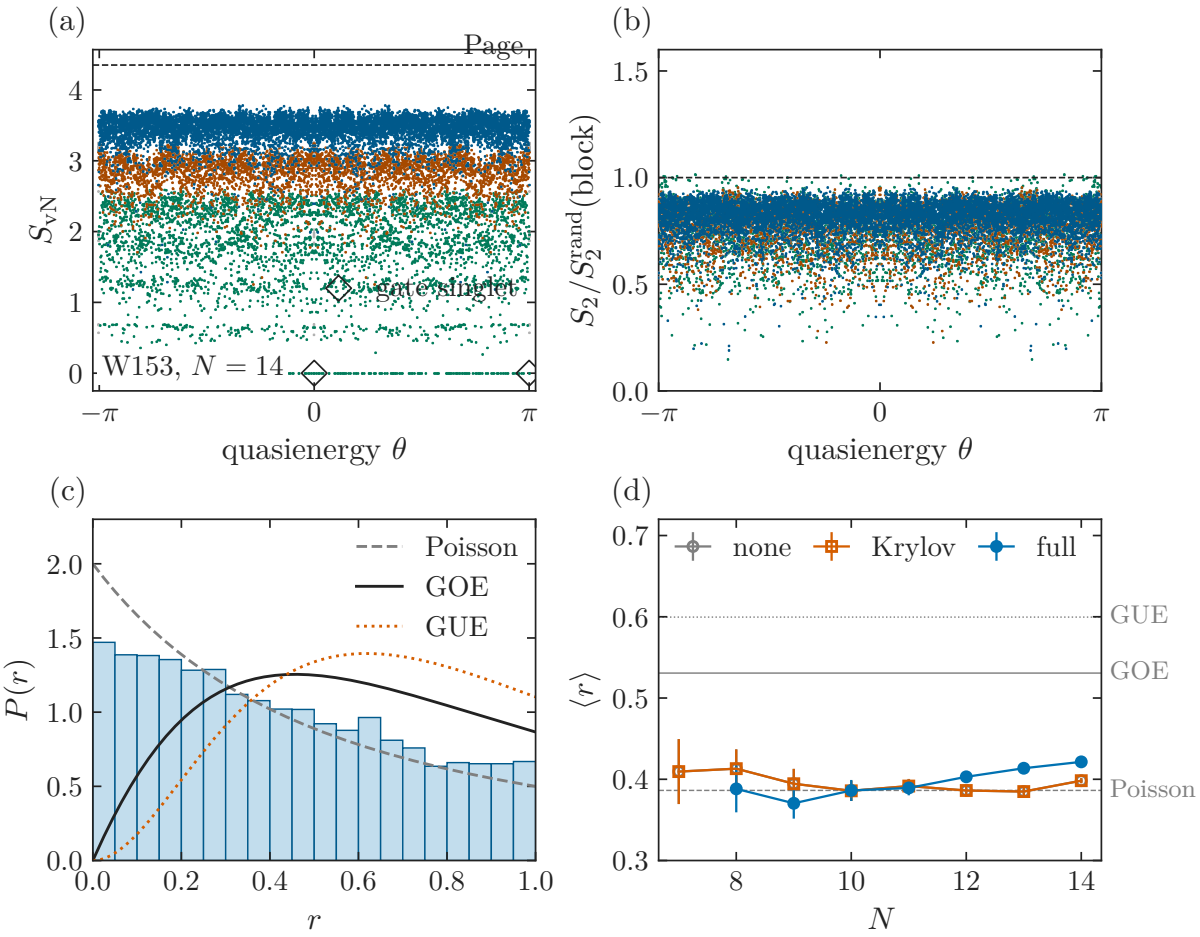


Figure 4: Rule 153 ($k = 2$, one-sided), largest N, obc0; panels as in Fig. 2. The entanglement bands in (a) are the blocks of the boundary-driven tower (sizes $2^{N-1}, 2^{N-2}, \ldots$). $P(r)$ is close to Poisson, while $\langle r \rangle$ drifts upward with N.

		sectors		blocks		obc		pbc $\langle r\rangle$		singlets	
k	W	obc	pbc	obc	pbc	$\langle r\rangle_{\text{full}}$	real j	complex j	naive ref.	obc	pbc
0	204	16384	16384	16384	16384	–	–	–	COE	0	0
1	108	3329	2627	5694	4517	0.383	0.404	0.386	COE	2070	1667
1	156	987	844	5740	5072	–	–	–	CUE	2439	2016
1	198	987	844	5740	5072	–	–	–	CUE	2439	2016
1	201	1897	2627	4143	4517	0.395	0.403	0.452	COE	1760	1667
2 one-sided	60	15	2	105	9	0.395	0.414	0.511	CUE	24	1
2 one-sided	102	15	2	105	9	0.395	0.414	0.511	CUE	24	1
2 one-sided	153	1	2	15	9	0.421	0.428	0.531	CUE	2	1
2 one-sided	195	1	2	15	9	0.421	0.427	0.531	CUE	2	1
2 XOR	105	8	7	15	20	0.356	0.405	0.391	COE	7	7
2 XOR	150	8	10	15	52	0.419	0.424	0.440	COE	7	6
3	54	2	2	3	9	0.523	0.383	0.483	COE	1	1
3	57	1	1	3	8	0.531	0.549	0.602	CUE	2	1
3	99	1	1	3	8	0.531	0.549	0.602	CUE	2	1
3	147	1	2	5	9	0.527	0.424	0.518	COE	4	1
4	51	1	1	16384	16384	–	–	–	COE	16384	16384

Table 3: Open chain against ring at $N = 14$, all sixteen rules. "naive ref." is the expectation for the complex sectors from the identified symmetries (mirror $\Rightarrow$ COE). The one-sided rules violate it (Table 4). The $N = 12$ table is `tab_r32_bc_H_N12.tex`.

($\chi^2 = 47$, against 1083 for $\beta = 1$). This is the textbook outcome for a chaotic system once its momentum sectors are complex. 147 (mirror) is COE-shaped and approaches 0.531 from below (0.469, 0.496, 0.518 for $N = 10, 12, 14$), and reaches 0.528 at $N = 16$ (block-first, Sec. 10). 54 (mirror) follows more slowly (0.458, 0.472, 0.483, and 0.510 at $N = 16$, now COE-shaped). On the open chain, 54 also lagged the other three rules by about two sizes.

One-sided rules: the tower is a boundary effect, and on the ring they are chaotic. On the ring the one-sided rules have 2 Krylov sectors and 9 blocks, against a tower of $N + 1$ blocks (60: 105 blocks at $N = 14$). In the complex sectors their statistics are $\beta = 1$ in *shape*:

- 153 and 195: $\langle r\rangle = 0.531$, $P(r < 0.15) = 0.063$, and $\chi^2_{\beta=1} = 27$ against 385 for the best Poisson+CUE mixture.
- 60 and 102: still rising (0.423, 0.471, 0.511, and 0.523 at $N = 16$), and closer to COE than to any mixture.

This contradicts the naive reference: these rules are *not* mirror symmetric, so one expects CUE. There must therefore be an antiunitary symmetry inside each momentum sector. It is identified in Sec. 10: a bond reflection dressed by a rotation of one sublattice into the Hadamard eigenbasis. A search at $N = 8$ found nothing beyond the known layer swap $T_1 U T_1^{-1} = U^{-1}$, which preserves j and so gives no time reversal inside a sector. It looked for W with $WUW^{-1} = U^{\pm 1}$, and separately with $W\Omega_T W^{-1} = \Omega_T^{-1}$, over $W = D\,L\,R^a T_s$: $a \in \{0, 1\}$, all s, L the identity or a uniform or sublattice product of X or H (and of Z without D), and D an *arbitrary* diagonal phase operator, solved exactly as a consistency problem on the support graph. For 147 the search returns the reflection, which validates it. This first search failed because the dressing rotation maps the computational basis to the Hadamard eigenbasis (a rotation by $\pi/8$), which is not among the single-site operators tried; the same search in the mixed frame finds it.

The consequence for the classification is that **the one-sided class is defined by its boundary**. With open ends the tower splits the space into $N + 1$ blocks and the statistics stay near Poisson. On the ring the same bulk dynamics is chaotic.

$k = 2$ **XOR: still arithmetic.** The ring makes the spectra of 150 and 105 *more* degenerate. At $N = 14$, 150 has 255 distinct quasienergies and 105 has 251, against $2187 = 3^7$ for both on

k	W	mirror	$\#r$	$\langle r\rangle$	$P(r<0.15)$	$\chi^2_{\rm P}$	$\chi^2_{\beta=1}$	$\chi^2_{\beta=2}$	$\chi^2_{{\rm P}+\beta=2}$
3	57	no	13944	0.602	0.023	8482	1083	47	47
3	99	no	13944	0.602	0.023	8482	1083	47	47
3	147	yes	13938	0.518	0.082	3444	124	4324	400
3	54	yes	13938	0.483	0.105	2119	850	9869	466
2 one-sided	60	no	13938	0.511	0.079	3228	163	4873	399
2 one-sided	102	no	13938	0.511	0.079	3228	163	4873	399
2 one-sided	153	no	13938	0.531	0.063	4191	27	2458	385
2 one-sided	195	no	13938	0.531	0.063	4191	27	2458	385
2 XOR	150	yes	4914	0.440	0.221	316	3656	23058	150
2 XOR	105	yes	1944	0.391	0.453	1043	4811	23681	1036

Table 4: Shape of $P(r)$ in the complex momentum sectors of the ring, $N = 14$: χ^2 of a 10-bin histogram (9 degrees of freedom) against the Atas surmises, and against the best Poisson+CUE mixture. A crossover from Poisson towards CUE can pass through the COE *mean*, but then its small-r tail is too heavy ($P(r) \sim r^\beta$). The surmise values of $P(r < 0.15)$ are 0.261 (Poisson), 0.064 ($\beta = 1$) and 0.019 ($\beta = 2$). The surmise deviates from large-N random matrices by about 1%, so χ^2 of a few tens at $n \approx 1.4 \times 10^4$ is a good fit.

the open chain. Their statistics are not those of a chaotic spectrum either: $\langle r\rangle = 0.440$ (150) and 0.391 (105) in the complex sectors, with a small-r tail of $P(r < 0.15) = 0.22$ and 0.45 (Poisson: 0.26; $\beta = 1$: 0.06). That tail is the signature of near-coincident levels in an arithmetic spectrum (Table 4). The spectra are arithmetic, not chaotic, as on the open chain.

$k = 1$: still exponentially fragmented. At $N = 14$ the ring has 2627 sectors for 108 and 201 (open chain: 3329 and 1897) and 844 for 156 and 198 (open: 987). The blocks stay small: the largest has 143 states for 108/201 and 25 for 156/198. On the ring, 108 and 201 have *identical* block structures (sectors, blocks, singlets and the count of distinct levels), but different statistics (0.399 and 0.417, from 2547 ratios each). This is consistent with a relabelling of the basis that maps one rule's sector graph onto the other's, while the gate H is not invariant under it. The same holds for 156 and 198, which are mirror images. Within the large uncertainty of such small blocks, the statistics stay between Poisson and the structured value of the open chain.

Exact singlets on the ring. Every $k = 3$ and one-sided rule has exactly *one* gate singlet on the ring, at $\theta = 0$, for every N (the XOR rules keep about one per sector, as on the open chain). The singlets at $\theta = \pi$ of the open chain are gone, so they are boundary states. For 57, 99, 147, 153 and 195 the ring singlet is a product state ($S_{\rm vN} = 0$). For 54, and also for 60 and 102, it is entangled, with the *same* entropy as the entangled singlet of 54 on the open chain: $S_{\rm vN} = 0.4045,\ 0.3447,\ 0.2908,\ 0.2430$ for $N = 8, 10, 12, 14$. It is one and the same state (overlap 1 at $N = 10$): $|h_+\rangle^{\otimes N}$ projected off the frozen vacuum, with its entropy in closed form (Sec. 10).

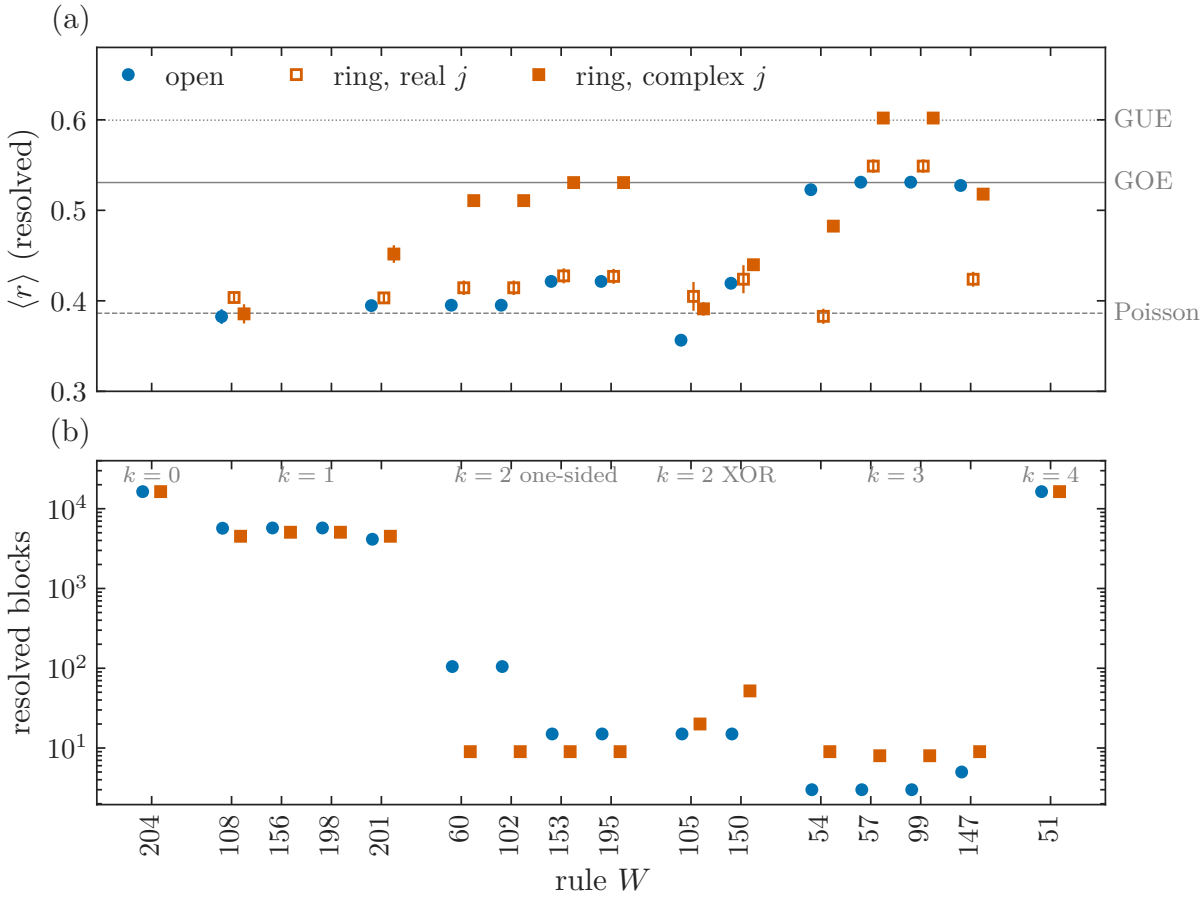


Figure 5: Open chain against ring, each rule at the largest even N available for both boundaries ($N = 14$ for the $k \geq 2$ non-arithmetic rules, $N = 12$ otherwise). (a) Resolved $\langle r \rangle$. For the ring, the real momentum sectors (open squares) and the complex sectors (filled squares) are shown separately. (b) Number of resolved blocks.

7.3 What survives

class	open chain	ring
$k = 3$	chaotic (COE)	chaotic: CUE (57, 99) or COE (147, 54), as the symmetry dictates
$k = 2$ one-sided	boundary tower, intermediate ($\langle r \rangle \approx 0.424$, flat)	tower gone, chaotic (COE, dressed-reflection time reversal)
$k = 2$ XOR	arithmetic	arithmetic, even more degenerate
$k = 1$	exponential fragmentation	unchanged
$k = 0, 4$	trivial	trivial

The $k = 3$, XOR, $k = 1$ and trivial classes are robust. The one-sided class is not a bulk class: its non-ergodicity comes from the open ends. "Chaos needs three firing patterns" is therefore a statement about the open chain. On the ring, two firing patterns suffice if they depend on a single neighbour.

Open points (status 2026-10-15).

1. *Resolved:* the antiunitary symmetry of the one-sided rules (Sec. 10).
2. *Resolved:* the identity of the entangled singlet shared by 54 and 60/102 (Sec. 10).

3. *Open, not needed for any conclusion:* resolving the residual $\mathbb{Z}_2$ in the real momentum sectors ($L_A R$ for the mirror rules, and the dressed reflection S for the one-sided rules). Both operators are now explicit, so this needs only one extra symmetry in the pipeline.

8 The X gate

With $V = X$ every rule is a classical reversible cellular automaton. U permutes the basis, so its spectrum is fixed by the cycle structure alone: a cycle of length L contributes the L-th roots of unity, once each. The eigenvectors are the Fourier modes of the cycles,

$$|q, c\rangle = L^{-1/2} \sum_{t=0}^{L-1} e^{2\pi i q t/L}\, U^t |x_c\rangle.$$

Level statistics therefore measure orbit arithmetic, not chaos, and are not reported. The informative quantities are the cycle-length distribution and the entanglement of the Fourier modes. A mode is supported on L basis states, so its Schmidt rank is at most L and $S_{\mathrm{vN}} \leq \ln L$: short cycles force low entanglement.

Data and checks. All sixteen rules were run with the same Julia pipeline, on the open chain for $N = 6 \ldots 14$ and on rings for even $N = 6 \ldots 14$: 224 runs and 40 minutes in total, with no failure. For every run, the stored quasienergies equal the union of the roots of unity over the cycles to 2×10^{-14}. The cycles were computed independently in Python from the permutation (`spectral.xgate.cycle_lengths`). A test checks this against the dense permutation matrix at $N = 8$, for both boundaries. The cycle growth in Fig. 6 was computed in Python up to $N = 18$.

8.1 What the sixteen automata are, classically

Writing m, n for the left and right neighbour, the X-gate update is $x_i \to x_i \oplus f(m, n)$, with f the firing pattern:

- **Linear or affine** (f of degree one): the XOR rules 150 ($f = m \oplus n$) and 105 ($1 \oplus m \oplus n$), and all one-sided rules (60: $f = m$; 102: $f = n$; 153: $1 \oplus n$; 195: $1 \oplus m$). These are solvable by linear algebra over $\mathbb{F}_2$. X-102 (flip iff the right neighbour is 1) is exactly the deterministic Floquet quantum East model, whose quench dynamics are solved exactly [44]; 60 is its mirror image.
- **Known interacting integrable automata.** 54 fires iff at least one neighbour is 1: this is RCA54, the interacting integrable Rule 54 with solitons [33]. 201 fires iff both neighbours are 0: this is the Floquet-PXP automaton [34]. On the ring the global spin flip is a symmetry of the X automata. It maps 147 onto 54, 108 onto 201, 156 onto 198 and 57 onto 99. With H it is not a symmetry, and the obc0 boundary (ghost sites fixed to 0) breaks it too: for example, 108 and 201 have 4868 and 3492 cycles on the open chain at $N = 14$, but 4156 each on the ring.
- **Trivial**: 204 ($U = 1$) and 51 ($U = X^{\otimes N}$).
- **Nonlinear and not known to be integrable**: only 57 and 99 (fire unless $(m, n) = (0, 1)$, or $(1, 0)$).

8.2 Cycle structure: only 57/99 have exponentially long orbits

On the open chain, the longest cycle of 57/99 holds a *finite fraction* of the Hilbert space: 129 880 of 2^{18} states at $N = 18$. A random state lies on a cycle of average length 0.22 to 0.50 times 2^N

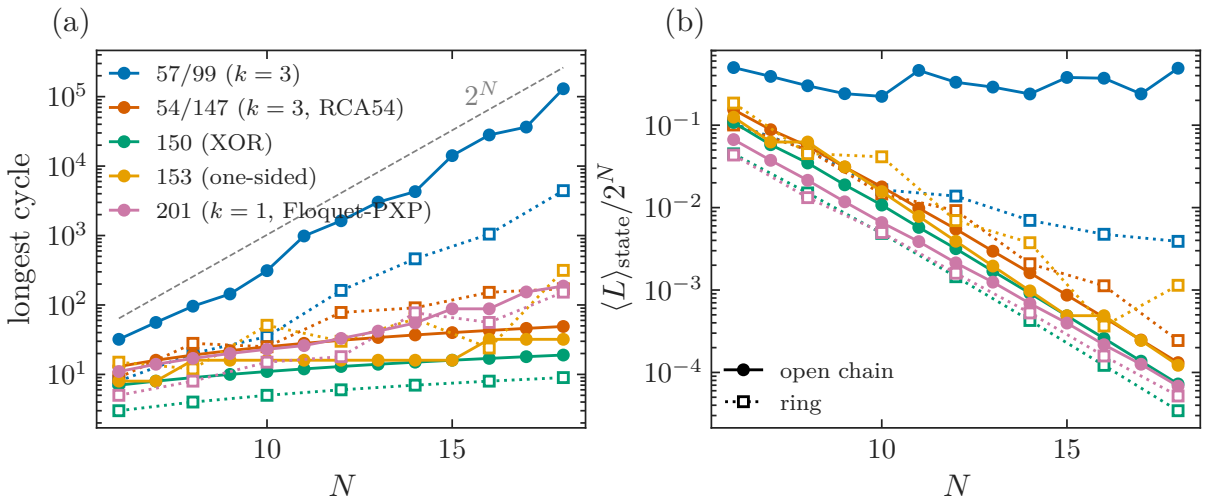


Figure 6: X gate: (a) the longest cycle and (b) the mean cycle length seen by a uniformly random basis state, $\langle L \rangle_{\text{state}} = \sum_c L_c^2/2^N$, over 2^N. Filled: open chain; open symbols: ring. Computed from the permutation up to $N = 18$.

for every N from 6 to 18. Classically, this is an ergodic orbit structure. Every other rule has orbits that grow at most linearly:

- RCA54: longest cycle $3N-5$ for 54 (147, which is RCA54 only on the ring: 26 at $N = 14$);
- XOR: $N + 1$;
- one-sided: 16 or 32, independent of N (a period that is a power of two, as for linear automata);
- Floquet-PXP (201, 108): slow growth, 187 states at $N = 18$.

On the ring, the cycles of 57/99 still grow exponentially, but more slowly (4428 states at $N = 18$), and their share of the space falls to $\langle L \rangle_{\text{state}}/2^N = 0.007$ at $N = 14$. The classically ergodic orbit of 57/99 on the open chain needs the boundary. The reason is a conservation law: on the ring X-57 conserves the number of domain walls, which the fixed-0 ends of the open chain create and destroy (Sec. 10). This is the mirror image of the one-sided rules with H, which need the ring to become chaotic.

8.3 Eigenstate entanglement: H against X at one N

Table 5 and Fig. 7 give the following picture:

- **57/99:** on the open chain, X and H are *equally* thermal (0.98 of Page). The cycle Fourier modes of an ergodic orbit look like random states. On the ring, the X value drops to 0.69 with the shorter cycles, while H stays at 0.99.
- **54/147 (RCA54):** X gives 0.70, bounded by its short orbits ($\ln 37 = 3.6$ against Page 4.35). H gives 0.89 and 0.93, and its spectrum is GOE (Sec. 6). **The Hadamard turns an interacting integrable classical automaton into a chaotic quantum circuit.**
- **One-sided:** 0.44 to 0.47 (X) against 0.47 to 0.66 (H) on the open chain. On the ring both rise (X: 0.84; H: 0.89 to 0.92), but only H is chaotic there; X stays affine, with period at most a few hundred.
- **XOR:** X (0.59) is slightly *above* H (0.52) on the open chain; both are arithmetic.

		open chain					ring				
k	W	cycles	longest	$\langle L\rangle/2^N$	$\bar{S}_X$	$\bar{S}_H$	cycles	longest	$\langle L\rangle/2^N$	$\bar{S}_X$	$\bar{S}_H$
0	204	16384	1	6.1e-05	0.00	0.00	16384	1	6.1e-05	0.00	0.00
1	108	4868	33	4.3e-04	0.11	0.06	4156	77	5.3e-04	0.17	0.12
1	156	2280	60	7.9e-04	0.14	0.02	2202	35	6.5e-04	0.19	0.04
1	198	2280	60	7.9e-04	0.14	0.02	2202	35	6.5e-04	0.19	0.04
1	201	3492	55	6.9e-04	0.14	0.07	4156	77	5.3e-04	0.17	0.11
2 one-sided	60	1044	16	9.7e-04	0.44	0.47	300	63	3.8e-03	0.84	0.89
2 one-sided	102	1044	16	9.7e-04	0.44	0.47	300	63	3.8e-03	0.84	0.89
2 one-sided	153	1024	16	9.8e-04	0.47	0.66	300	63	3.8e-03	0.84	0.92
2 one-sided	195	1024	16	9.8e-04	0.47	0.66	300	63	3.8e-03	0.84	0.92
2 XOR	105	1096	15	9.1e-04	0.59	0.52	1172	14	8.5e-04	0.58	0.75
2 XOR	150	1096	15	9.1e-04	0.59	0.51	2344	7	4.3e-04	0.43	0.57
3	54	632	37	1.6e-03	0.71	0.89	732	91	2.1e-03	0.65	0.82
3	57	34	4288	2.4e-01	0.98	0.98	500	462	7.0e-03	0.69	0.99
3	99	34	4288	2.4e-01	0.98	0.98	500	462	7.0e-03	0.69	0.99
3	147	632	26	1.6e-03	0.70	0.93	732	91	2.1e-03	0.65	0.93
4	51	8192	2	1.2e-04	0.16	0.00	8192	2	1.2e-04	0.16	0.00

Table 5: X gate at $N = 14$, both boundaries: number of cycles, the longest, $\langle L\rangle_{\text{state}}/2^N$, and the mean half-chain entanglement of all eigenstates over the Page value ($\bar{S}_X$). $\bar{S}_H$ is the same mean for the H gate, over its non-degenerate eigenstates (degenerate ones have basis-dependent entanglement).

- $k = 1$**:** both are low (X 0.11 to 0.14; H 0.02 to 0.07). These are fragmented rooms, whatever the gate.

8.4 What the comparison says

1. Quantum chaos with H does *not* require classical chaos with X. Two of the four chaotic $k = 3$ rules, 54 and 147, are the interacting integrable RCA54 with X, yet GOE-chaotic with H. On the ring, 57/99 keeps its CUE chaos with H, although its X orbits no longer fill a finite share of the space. What the chaotic rules share is the firing count $k = 3$, not the classical dynamics. Deforming classical automata by non-commuting gates is the setting of Ref. [32]; for these rules, the deformation by H yields chaos.

2. Conversely, the only classically ergodic automaton (57/99 on the open chain) is also chaotic with H.

3. With X, every rule splits into cycles, so "fragmentation" is total and carries no information. With H, the fragmentation is a property of the rule class: exponential at $k = 1$ on both boundaries, and boundary-driven for the one-sided rules.

9 Thermalisation beyond the gap ratio

The gap ratio is the most robust single diagnostic, but it probes only nearest-neighbour level correlations of whole symmetry blocks. Three complementary diagnostics were added:

- the *spectral form factor* (SFF), which measures long-range spectral correlations and the time scale of ergodicity;
- *ETH for a local observable*, i.e. whether the individual eigenstates are thermal;
- *quench dynamics* from product states, i.e. whether specific initial states thermalise.

Thermal means infinite temperature inside the block a state lives in.

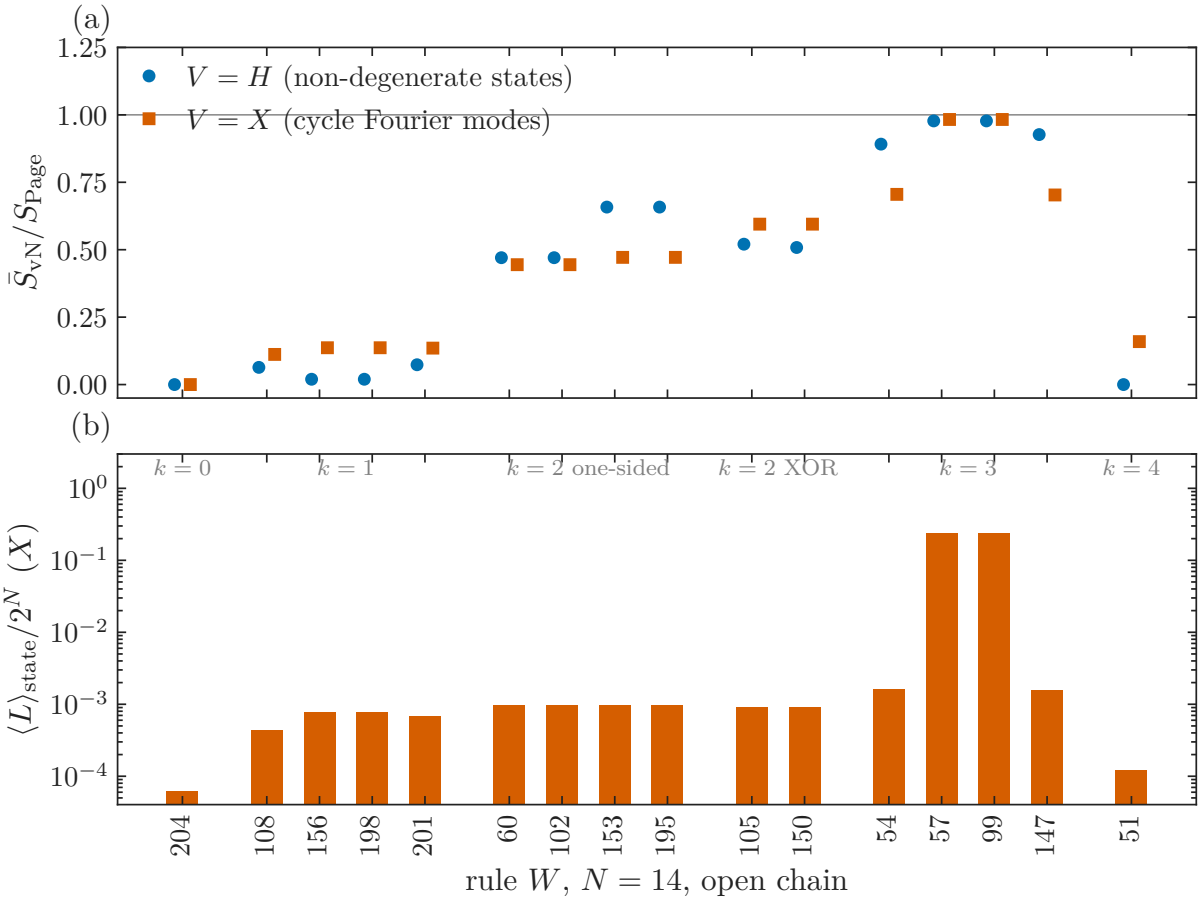


Figure 7: H against X at $N = 14$, open chain. (a) Mean eigenstate entanglement over the Page value. (b) X gate: $\langle L \rangle_{\mathrm{state}}/2^{N}$.

9.1 Spectral form factor

The SFF $K(t) = \langle |\mathrm{Tr}_b U^t|^2 \rangle / d$ was computed over the complex momentum sectors on the ring. There each block's eigenphases fill the circle and the reference is COE or CUE (Sec. 7). On the open chain every block is real, with a $\pm\theta$-symmetric spectrum, so the standard form factors do not apply. A single circuit has no ensemble, and the SFF does not self-average, so the curves were averaged over the sectors and over logarithmic time bins. At $t \gtrsim 100$ each bin pools hundreds of points, so the residual noise is a few percent.

Fig. 8(b) shows two systematic features:

1. A non-universal excess at short times ($t \approx 5 \ldots 30$ periods): at $N = 16$ about 10^2 times the ramp for 57 and 153, and 10^3 to 10^4 for 54, 60 and 147. It grows with N for every rule, most strongly for 147 and 60.

2. A slow approach to the ramp. For 57, $K/K_{\mathrm{RMT}} \approx 1.4$–$1.6$ persists over two decades and reaches 1.1 only near the Heisenberg time D.

A sharp Thouless time (agreement within $\pm 30\%$) is therefore not reached well before D at these sizes. The robust proxy t_2 ranks the rules clearly (Fig. 8(c)). For 57 it grows slowly, from 25 to 78 periods for $N = 10 \ldots 16$ (roughly as $D^{0.3}$). For the other chaotic rules it is 5 to 13 times longer at $N = 16$ and grows faster. The early excess suggests that part of the space of these rules has approximately periodic dynamics. This is a tentative reading; it fits the slow scar-defect dynamics below.

k	W	ring $\langle r\rangle$ (N)	ETH exponent	t_2 ($N=16$)	random product open	random product ring	scar $+\,\lvert h_-\rangle$ open	scar $+\,\lvert h_-\rangle$ ring
3	57	0.598 (16)	-0.40	78	1.00	1.00	0.95	0.96
3	54	0.510 (16)	-0.13	990	0.99	0.99	0.31	0.44
3	147	0.528 (16)	-0.27	426	1.00	1.00	0.63	0.60
2 one-sided	153	0.527 (16)	-0.24	564	0.91	1.00	0.00	0.89
2 one-sided	60	0.523 (16)	-0.17	747	0.72	0.99	0.00	0.78
2 XOR	150	0.431 (14)	–	–	0.56	0.69	0.26	0.42
1	201	0.452 (14)	–	–	0.08	0.14	0.29	0.31
1	156	–	–	–	0.00	0.01	0.11	0.14

Table 6: Thermalisation summary. Ring $\langle r\rangle$: complex momentum sectors at the largest N (in brackets). ETH exponent: slope of $\log\sigma$ against $\log d$ for the eigenstate-to-eigenstate fluctuation σ of the magnetisation density $m=\sum_i Z_i/N$ in the complex ring sectors ($N=8\ldots14$); ETH predicts $-1/2$. It is shown only for rules whose momentum sectors are single Krylov sectors. t_2: time after which the log-binned SFF stays within a factor 2 of the RMT ramp, at $N=16$. Last four columns: $S_{\mathrm{half}}/S_{\mathrm{Page}}$ averaged over $t=200\ldots300$ after a quench at $N=14$, from random computational product states (average of 8) and from $\lvert h_+\rangle^{\otimes N}$ with one $\lvert h_-\rangle$ in the middle.

9.2 ETH for a local observable

The diagonal matrix elements $\langle n|m|n\rangle$ of the magnetisation density were computed in all non-degenerate eigenstates of the complex ring sectors ($N=8\ldots14$). On the open chain the central Z was used in the largest gate-algebra block, at even N; this gives the same ordering. For 57 the fluctuation σ falls as $d^{-0.40}$ (open chain: $d^{-0.36}$), close to the ETH value $d^{-1/2}$ at these sizes. The other chaotic rules converge markedly more slowly: -0.27 (147), -0.24 (153), -0.17 (60) and -0.13 (54). This is the same ordering as their $\langle r\rangle$ and SFF convergence. For 150 nearly all eigenstates are degenerate (free fermions), so a diagonal ETH test is meaningless there. The $k=1$ momentum sectors contain many Krylov sectors, so the test does not apply to them either.

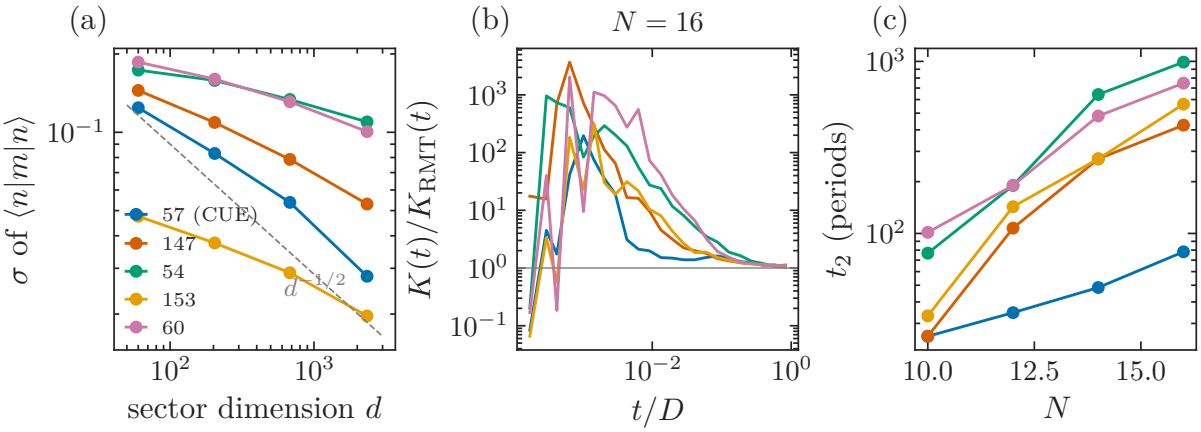


Figure 8: (a) ETH: eigenstate-to-eigenstate fluctuation of the magnetisation density in the complex ring sectors, against the sector dimension; dashed: $d^{-1/2}$. (b) SFF over the RMT ramp at $N=16$, log-binned. (c) t_2 against N. Colours as in (a).

9.3 Quench dynamics

Product states were evolved with the sparse layers at $N=14$ for 300 periods (Fig. 9, Table 6). The findings, class by class:

- $k = 3$*:* random product states reach the Page value on both boundaries (0.99 to 1.00). For 57 so do Néel and the scar with one defect.
- *One-sided, a boundary effect in real time:* random states saturate at 0.72 (60) and 0.91 (153) of Page on the open chain, but at 0.99 to 1.00 on the ring.
- *XOR (150):* stays well below Page (0.56 to 0.69). Néel does not move at all on the ring.
- $k = 1$*:* hardly entangles (201: 0.08 to 0.14, 156: about 0); this is the fragmentation.
- *The scar:* $|h_+\rangle^{\otimes N}$ returns with probability 1 for every rule.
- *One* $|h_-\rangle$ *defect on top of the scar:* slow for 54 and 147. Their entanglement is still rising at $t = 300$ (0.3 to 0.6 of Page), while for 57 it saturates within about 100 periods. This is a bulk analogue of the edge mode of 57: a localised excitation on the inert $|h_+\rangle$ vacuum that leaks into the bulk only slowly. On the open chain the one-sided rules keep the defect state at exactly zero half-chain entanglement. Everything to the left of the defect is the inert $|h_+\rangle$ part of the tower, so the cut sees a product state.

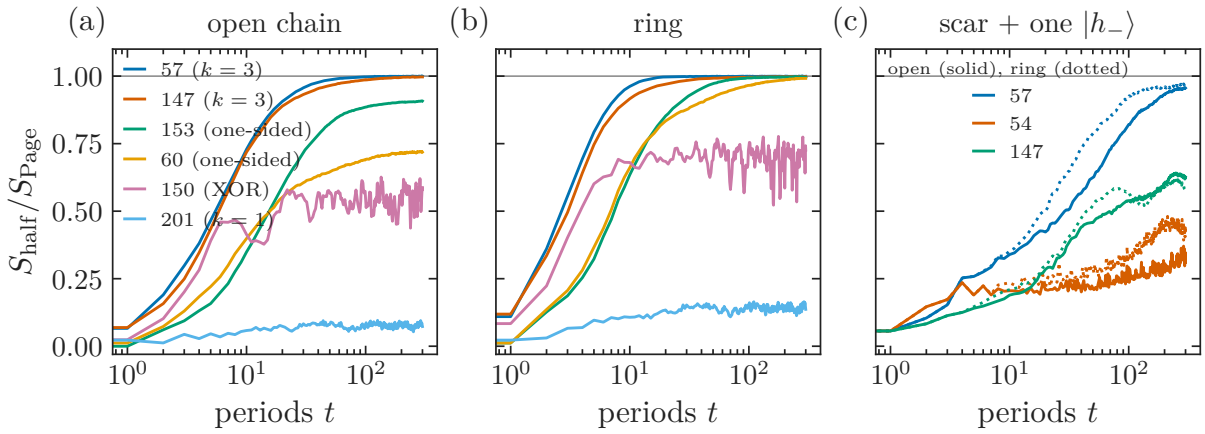


Figure 9: Half-chain entanglement over the Page value after quenches, $N = 14$. (a) Random computational product states, open chain (average of 8); (b) the same on the ring; (c) $|h_+\rangle^{\otimes N}$ with one $|h_-\rangle$ in the middle, for the $k = 3$ rules (solid: open chain, dotted: ring).

Verdict. The ordering of all diagnostics is the same: 57/99 are the fastest and cleanest thermalisers (CUE/COE statistics, ETH-like fluctuations, short t_2). Behind them come 153, 147, 60 and 54, all chaotic on the ring, with the convergence slowing in that order. XOR and $k = 1$ do not thermalise. The non-thermal exceptions inside the chaotic rules are all tied to the inert $|h_+\rangle$ vacuum: the exact scars, the edge mode of 57, and the slow defect states of 54 and 147.

10 Exact structures behind the classes

This section collects the exact structures found in the follow-up (2026-10-15), which explain the classes: the flip map, the hidden time reversal of the one-sided rules, the universal scar, the kink dispersion of 156/198 and the domain-wall charge of X-57. It also reports the block-first runs beyond the dense-sector limit (step 9), the eigenvector analyses at small N (step 10), and the checks of the assumptions those runs make. Every statement is pinned by a test in `code/qca_fragmentation/tests/test_spectral.py`.

10.1 Flip partners: the X flip turns H into XHX

The physical spin flip is $X^{\otimes N}$. It maps a rule W onto its partner W' (the firing pattern $f(m,n) \to f(1-m,1-n)$), but the gate changes with it:

$$X^{\otimes N}\, U_W(H)\, X^{\otimes N} = U_{W'}(XHX), \qquad XHX = \tfrac{1}{\sqrt{2}}(X - Z) = -ZHZ. \tag{5}$$

Because $Z^{\otimes N}$ commutes with the controls, $Z^{\otimes N} U_{W'}(G) Z^{\otimes N} = U_{W'}(ZGZ)$. So $U_W(H) \cong U_{W'}(-H)$ up to the basis change $Z^{\otimes N}$: the partner rule with a -1 on every firing. This is a conditional phase, not a global one. Verified as operators at $N = 8$, on the ring, for the pairs 54/147, 108/201, 60/195, 102/153 and 57/99. The consequences are:

- On the ring, flip partners have *identical* supports, hence identical Krylov sectors and blocks, but *different* spectra. Examples: 108 and 201 ($\langle r \rangle = 0.399$ and 0.417 from identical block structures), 60/102 and 153/195 (0.511 and 0.531), and 54 and 147 (0.483 and 0.518).
- The obc0 boundary (ghost sites fixed to 0) breaks the flip already for X. With X on the ring, 108 and 201 have 4156 cycles each; on the open chain they have 4868 and 3492.
- 57/99 and 156/198 are also *reflection* partners, and the reflection is an exact symmetry with either gate. Their spectra are identical in every dataset.

10.2 The hidden symmetry of the one-sided rules

On the ring, the one-sided rules have $\beta = 1$ statistics without mirror symmetry (Sec. 7). The time-reversal symmetry behind this is now identified:

$$S = B_e\, D\, R_{\text{bond}}\, B_e^{\mathsf{T}}, \qquad A = K\, L_A\, S, \tag{6}$$

where:

- R_{bond} is the bond-centred reflection $i \to 1 - i$, which exchanges the two sublattices;
- B_e rotates the even sites into the eigenbasis $\{|h_+\rangle, |h_-\rangle\}$ of the Hadamard;
- $D = X^{\otimes N}$ for 153/195, and a diagonal sign operator for 60/102.

Verified as explicit operators at $N = 8$ and 10, for all four rules, to 10^{-13}:

- S is real, $[S, U] = 0$, and $S T_2 S^{-1} = T_2^{-1}$ (S flips the momentum);
- A preserves the momentum and maps $U \to U^{-1}$ and $\Omega_T \to \Omega_T^{-1}$;
- $A^2 = +1$: COE, not Kramers pairs.

In the mixed frame (Hadamard basis on the targets of one layer, computational basis on its controls) that layer is diagonal, and S becomes a dressed reflection. This is why a search in the computational basis could not find it. The single-site rotation needed is $V = (|h_+\rangle, |h_-\rangle)$, a rotation by $\pi/8$, not H. For 57/99 the same search returns nothing, consistent with CUE.

Correction to Sec. 7. In the real momentum sectors ($j = 0,\ N/4$), S preserves j and commutes with U. So the real sectors of the one-sided rules carry an unresolved $\mathbb{Z}_2$, exactly as the reflection does for the mirror rules. Their real-sector values (0.414 and 0.428 at $N = 14$) mix two symmetry sectors and are not diagnostic. The conclusions used only the complex sectors, so they are unchanged.

10.3 Scars: one universal eigenstate, edge-projected

$|h_+\rangle^{\otimes N}$ is a $+1$ eigenvector of every controlled-Hadamard gate, for every rule and boundary, because $H|h_+\rangle = |h_+\rangle$ whatever the controls read. All gate singlets of the chaotic rules follow from it (found from the gate algebra at $N \leq 12$ and identified site by site):

- $\theta = 0$: $|h_+\rangle^{\otimes N}$ itself.
- *Rules with a frozen vacuum* (54 on both boundaries, 60/102 on the ring): its projection off the vacuum, $|\psi\rangle \propto |h_+\rangle^{\otimes N} - \cos^N(\pi/8)\,|0\ldots0\rangle$. This is the "entangled singlet". It has Schmidt rank 2, its entanglement follows in closed form and matches the data to 10^{-6} for $N = 6\ldots14$, and it falls to zero as $\cos^N(\pi/8) \to 0$. It is the same state in all three rules (overlap 1).
- *Boundary singlets* ($\theta = \pi$, or 0 with two of them): $|h_-\rangle$ on an end site whose gate fires for both values of the inner neighbour. With the ghost fixed to 0, that site picks up -1 per period deterministically. This predicts the counts exactly: 57 (right end) 2, 99 (left end) 2, 147 (both ends) 4, 54 (neither end, vacuum frozen) 1. On the ring there are no ends, hence exactly one singlet.

These are exact scars, but of the simplest kind: a product eigenstate of the gate (or its projection) embedded in a chaotic spectrum, in the spirit of the projector embedding [20].

The outliers of 57. These are *not* singlets. They are an *edge mode on the $|h_+\rangle^{\otimes N}$ vacuum*: away from the left end the state is $|h_+\rangle$ on every site (overlap rising from 0.86 to 0.93 into the bulk), and the disturbance is localised on the first four sites. The two quasienergies are one mode: $\pm0.3122\pi$ alone, and $\pm0.6877\pi = \pm0.3122\pi + \pi$ together with the right-end $|h_-\rangle$ singlet. The mode lives at the left end because there 57's gate depends on the inner neighbour (it fires on 00, not on 01); the right end is inert. On the ring the strongly non-thermal states disappear (minimum η 0.41 to 0.55, against 0.13 to 0.27 on the open chain).

10.4 156/198: one kink per segment, with an exact dispersion

For 156 (fires only on left = 1, right = 0), a 01 domain wall is frozen and a 10 wall is mobile. Between two frozen walls a segment of L free sites holds exactly one kink, in $L+1$ positions. No gate couples two segments, so $U = \bigotimes_s U_s$. The single-kink spectrum is

$$\cos\theta_m = -\sin^2\frac{\pi m}{L+1}, \quad m = 1\ldots\lfloor(L+1)/2\rfloor \ (\pm \text{ pairs; } \theta=\pi \text{ at } m=\tfrac{L+1}{2}), \qquad \text{plus } \theta = 0, \tag{7}$$

i.e. $\cos\theta(q) = (\cos q - 1)/2$ with $q = 2\pi m/(L+1)$. It does not depend on the brickwork offset, and it reproduces the exact spectra of the largest Krylov sectors at $N = 12$ as sums over their segments (including the end segments). Blocks have at most about 125 states, so their levels are sums of a handful of values from this small set. That gives many exact and near coincidences, hence $\langle r\rangle \approx 0.42$: neither independent levels nor level repulsion.

10.5 X-57: the ergodic orbits need the boundary

On the ring, X-57 (and X-99) *conserves the number of domain walls.* A search over sublattice-resolved 3- and 4-site densities finds exactly this one local charge (plus its 4-site extension) on the ring, and none on the open chain. There the ghost sites create and annihilate walls. Ring orbits are confined to fixed wall number, so their share of 2^N decays (0.007 at $N = 14$). On the open chain the boundary mixes the wall sectors and the orbits fill a finite fraction. For contrast, 54 and 150 keep local charges on both boundaries.

10.6 Beyond the dense-sector limit (block-first)

The production pipeline diagonalises whole Krylov sectors densely, so its memory is set by the largest *sector* (Sec. 11). For the cases that matter for the open questions, an exactly known symmetry was used to split the space *before* the dense step, by a sparse construction (`spectral/blockfirst.py`), and only eigenvalues were computed.

- *Ring,* $N = 16$ (57, 54, 147, 153, 60): T_2-momentum orbit bases first (≈ 8200 states per sector), then the half step Ω_T inside each sector. The eigenphases come from a Cayley transform, a Hermitian eigenvalue problem.
- *Open chain, one-sided rules:* the largest tower block, built directly in its product basis. For 153 it is $|h_-\rangle$ on site 0 with the other sites free ($N = 15$); for 60 it is site $0 = |1\rangle$ and site $N - 1 = |h_-\rangle$ ($N = 16$). Each has 16 384 states.

Validation against the production data: the ring code reproduces the Julia φ of every momentum sector to 10^{-14} ($N = 12$) and the per-block $\langle r\rangle$ to 10^{-11} ($N = 12$, 14). The tower blocks reproduce the Julia largest blocks level by level to 10^{-13} ($N = 10 \ldots 13$).

One pitfall, caught by that comparison: computing $|\theta|$ from arccos of the eigenvalues of $(U + U^{\mathsf{T}})/2$ smears exactly degenerate eigenvalues ± 1 by $\sqrt{\epsilon} \approx 10^{-8}$ into spurious near-duplicates. The real tower blocks therefore use the eigenvalues of U directly.

Results (Secs. 7 and 6):

- On the ring at $N = 16$: 57 is 0.5985 ± 0.0011 (CUE), 147 is 0.528 and 153 is 0.527 (COE-shaped), and 60 (0.523) and 54 (0.510) are still rising, now COE-shaped.
- On the open chain the one-sided statistics stop drifting: $\langle r\rangle \approx 0.424$ for 153 at $N = 15$ and for 60 at $N = 16$.

Checking the assumptions (user request). Block-first assumes that nothing splits the pieces further. This was tested at the production sizes with the generic-K trick:

- *Ring:* restricted to a momentum sector, K is a generic combination of T_2-invariant elements of the gate algebra (S_A, S_B, $\{S_A, S_B\}$, $U + U^\dagger$). One component per Krylov sector means that no hidden symmetry splits the sector: any such symmetry would also split this subalgebra.
- *Tower:* $K = \sum_i a_i g_i$ restricted to the block.

Results:

- In every tested complex momentum sector at $N = 16$ (57 j=1; 54, 153 j=1, 2; 147 j=1; 60 j=1, 3) the test gives exactly one component, equal to the single Krylov component. K is non-degenerate (gaps 4×10^{-6} to 2×10^{-4}), and the sector is invariant under Ω_T to 2×10^{-15}.
- The tower block of 153 at $N = 15$ is irreducible: one component of 16 384, gap 2.3×10^{-6}, and the gates leave the block invariant to 1.4×10^{-14}.
- The tower block of 60 at $N = 16$ is irreducible: one component of 16 384, gap 2.3×10^{-7}, and gate invariance 2×10^{-14}.

In the *real* momentum sectors the same test splits the space into two halves. This is how the residual $\mathbb{Z}_2$ of the one-sided rules was found, which led to their time reversal (above). The conclusions use only the complex sectors.

Eigenvectors at small N (step 10). The scar analyses use eigenvectors from the exact engine at $N \leq 12$:

- the gate singlets, from the one-dimensional gate-algebra blocks;
- the outliers of 57 and the ETH matrix elements, from the dense eigenvectors of U in the largest block.

The Julia pipeline was not changed, so no full recompute was needed.

11 Compute and memory

Budget and what was used. The task came with a budget of 8 hours of compute. The user later lifted it ("finish within the next 12 hours"). All runs used one 12th-generation i7 with 31 GB of memory, one job at a time.

- First night (v1 pipeline, 01:53 to 02:51): obc0, all 16 rules for $N = 6 \ldots 13$ and $N = 14$ for 9 rules; about 1.7 h including validation and memory tests.
- Continuation (v2 pipeline, from 12:32 on 2026-10-14): 485 successful runs and 7.1 h of wall time.

stage	runs	wall time
H, obc0 (rest of $N = 14$; $N = 15 \ldots 20$ by rule; even-N reruns of the mirror rules with φ)	117	3.55 h
H, rings $N = 6 \ldots 14$ (with Ω_T)	144	2.92 h
X, obc0 $N = 6 \ldots 14$	144	0.44 h
X, rings $N = 6 \ldots 14$	80	0.23 h

Every completed run exited with code 0. Runs cut off by the interruptions below were rerun, except rule 153 at $N = 15$. The 27 non-zero exit codes in the log are the 12:32 start of the runner, which split paths at the spaces in the repository path; those runs failed instantly and were rerun after the fix.

The most expensive runs are the single $\approx 16\,000$-dimensional sectors at $N = 14$: 15 to 18 minutes each, with a peak resident memory of 14 GB.

Interruptions.

1. *First night, 02:51*: the session's memory reaper stopped all jobs. The cause was the dense U of the chaotic rules: for these rules U is effectively dense, and storing it as a sparse matrix and running a component search over it cost 1.1 GB at $N = 12$, or about 18 GB at $N = 14$. Fixed in v2 (below); resumed with the user's go-ahead.
2. *About 20:08*: the machine shut down. The run lost was 54 at $N = 14$, whose old file had already been removed for the rerun with φ. The user then stopped the runs for a second reboot. The rerun was done after the restart.
3. *About 22:45*: the memory reaper stopped all jobs during rule 153 at $N = 15$ (above 17 GB). That run is the only one that never finished. It was not restarted (see the next paragraph).

Why the sizes stop where they do. The pipeline diagonalises each Krylov sector densely (the gate-algebra blocks are resolved inside it), so the memory is set by the largest *sector*. A 16 383-dimensional sector peaks at about 14 GB, and the memory grows faster than the square of the dimension (201: 2.6 GB at $N = 18$, sector 6765; 7.7 GB at $N = 19$, sector 10 946). The next sizes would need:

- 153/195 at $N = 15$ and 60/102 at $N = 16$: a sector of 2^{15}, about 50 GB;
- 150/105 at $N = 16$: about twice the 12 870-dimensional sector of $N = 15$, more than 30 GB;
- 201 at $N = 20$: a sector of 17 711 (F_{22}), about 20 to 23 GB, which is too close to the limit at which the session reaper stops every job.

These sizes were therefore not attempted, so every class is compared at the largest N that all its members share. That is $N = 14$ for $k = 3$ and for the one-sided rules, $N = 15$ for XOR, and $N = 19$ for $k = 1$ on the open chain; on the ring, $N = 14$ for every rule. Going further would need a block-first diagonalisation, which projects onto the gate-algebra blocks before the dense step. That is the obvious next optimisation.

The v2 pipeline (`generate_spectral_data_v2.jl`).

1. U is never formed: $Uv = L_B(L_A v)$ with the two sparse layers (a few MB at $N = 14$).
2. Krylov sectors come from a union-find over the supports of the *individual gates*. For $V = H$ this is the same partition as the components of $\mathrm{supp}(U)$ (R-T19 Prop. 4), and it is asserted against the old construction for every $N \leq 10$ run. For $V = X$, U is a permutation and is formed directly.
3. Symmetry checks are random-vector tests through the layers. Dense sector blocks are assembled column-chunk-wise, and the products inside each block are computed per batch of at most 2048 columns.

Validation: v2 reproduces v1 block by block (sector size, block size, sorted quasienergies, S_2^{Haar}, degeneracies) for seven rules at $N = 9$. Its ring spectra match the exact Python engine to $< 10^{-14}$, for both gates. Non-degenerate eigenvectors agree to 10^{-15} in S_{vN}; degenerate ones are basis dependent by construction.

Analysis cost. The first analysis pass looped over blocks with one boolean mask per block, which is $O(\text{blocks} \times \text{states})$: about 10^{11} operations for the $k = 1$ rules at $N = 20$. It now groups the states by a single sort (`stats.groups`). The full summary of 173 obc0 units takes 70 s, and its output is identical to the old one on every unit that was not rerun.

A Bibliography checks

Every entry below was verified on 2026-10-14 (second pass, Sec. 3.5: 2026-10-15) against the arXiv listing, the DOI or publisher page, or an institutional repository record. The journal references of Regnault–Nandkishore and Ponte *et al.* (PRL 114, 140401) were confirmed from the arXiv journal-ref field. Deutsch (1991) was confirmed through citing papers only (Rigol *et al.* 2008 and the reviews). Jordan (1875) is cited for the origin of the two-subspace decomposition; the modern statement is Halmos (1969).

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